

15. Magnetic Newton's Cradle

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***Repulsing, non-touching magnets** are used instead of colliding balls to make a new type of **Newton's cradle**. The new cradle can act in a similar way to a regular cradle, but can also exhibit other **interesting behaviour**. Explain and study the movement of this magnetic cradle.*



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Experiment

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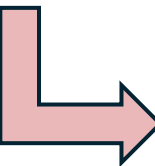


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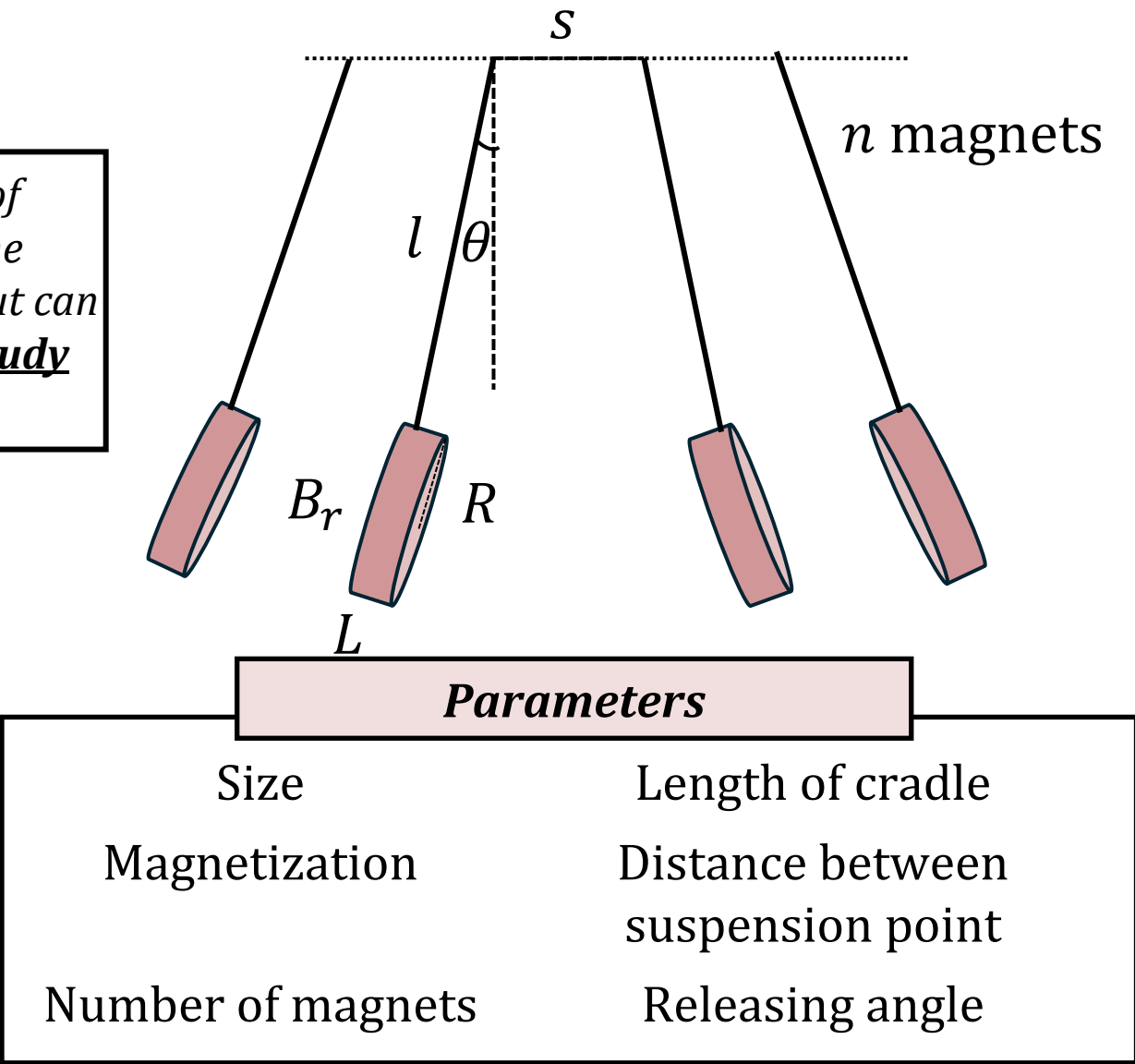
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Problem Analysis

Repulsing, non-touching magnets are used instead of colliding balls to make a new type of **Newton's cradle**. The new cradle can act in a similar way to a regular cradle, but can also exhibit other interesting behaviour. Explain and study the movement of this magnetic cradle.



Physical nature: a **coupled multi-body vibration system** driven by dynamic coupling of **magnetic and gravitational forces**



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Preliminary Experiment Phenomenon

Two Cradles

Parameters

$R=15\text{mm}$, $L=10\text{mm}$, $M=50\text{g}$, $B_r=1.38\text{T}$, $l=0.0892\text{m}$, $s=0.04\text{m}$, $\theta_0=44.3^\circ$



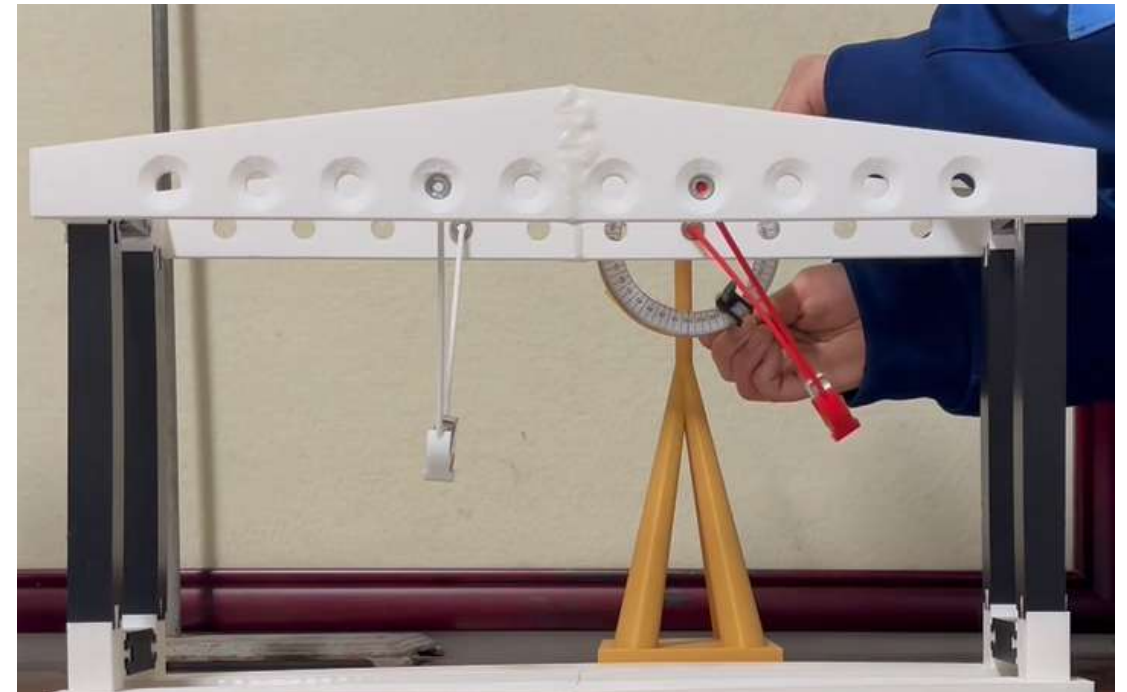
*Two cradles oscillate alternatively
--- periodic movement*



*NO direct collision
--- indirect “collision” due to
repulsive magnetic force*



Energy exchange continuously



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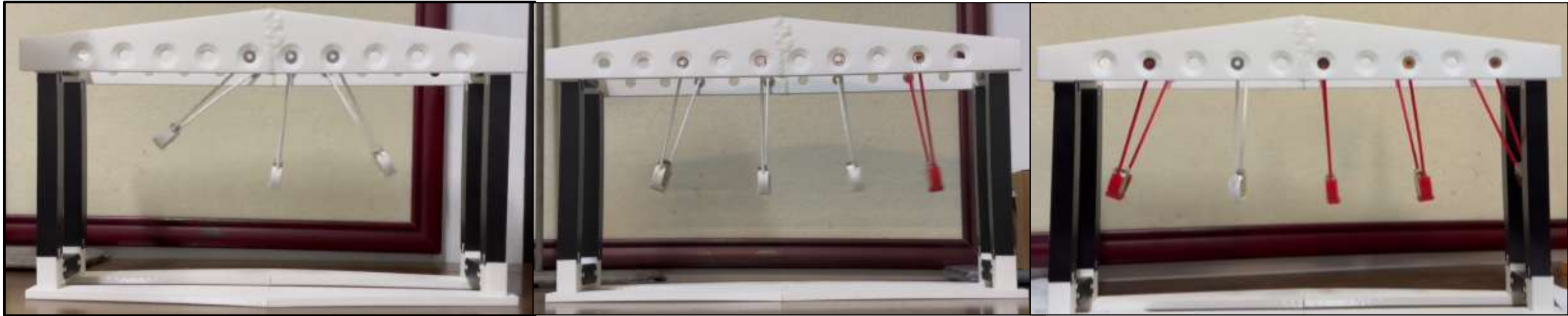
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Preliminary Experiment Phenomenon

More Cradles



- ① *Periodic Movement* ② *Collision Latency* ③ *Energy exchange*

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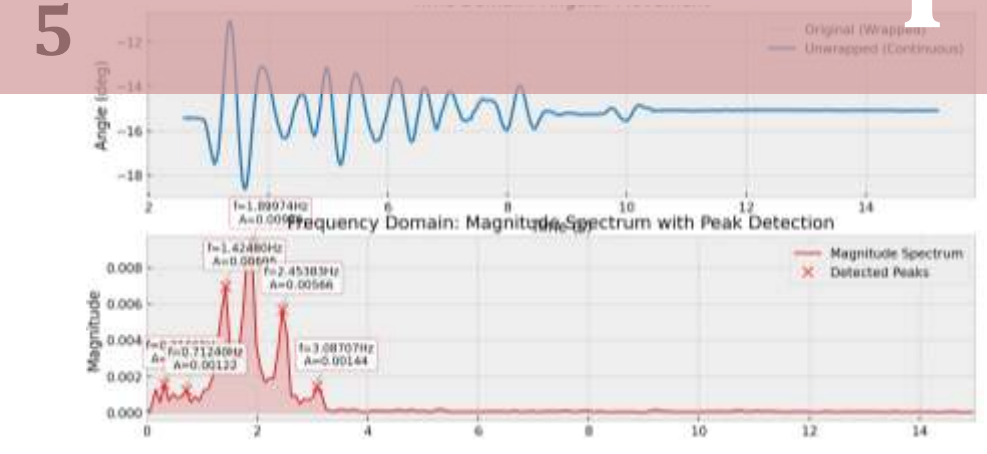
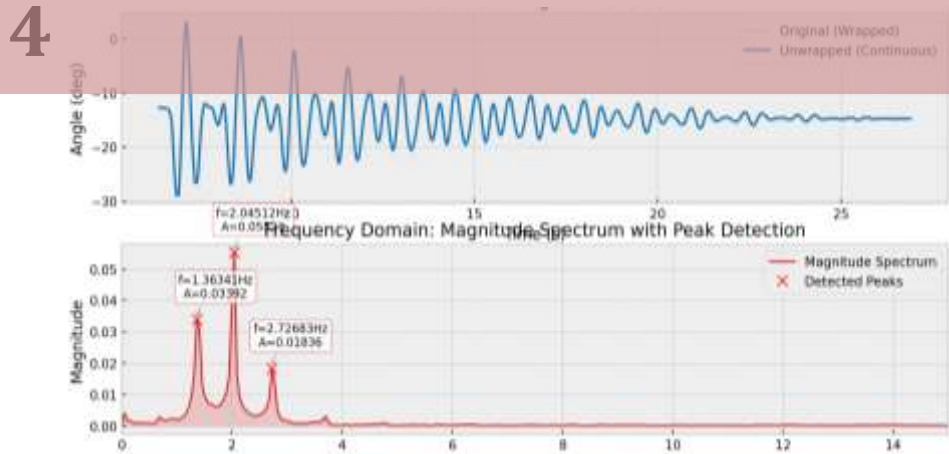
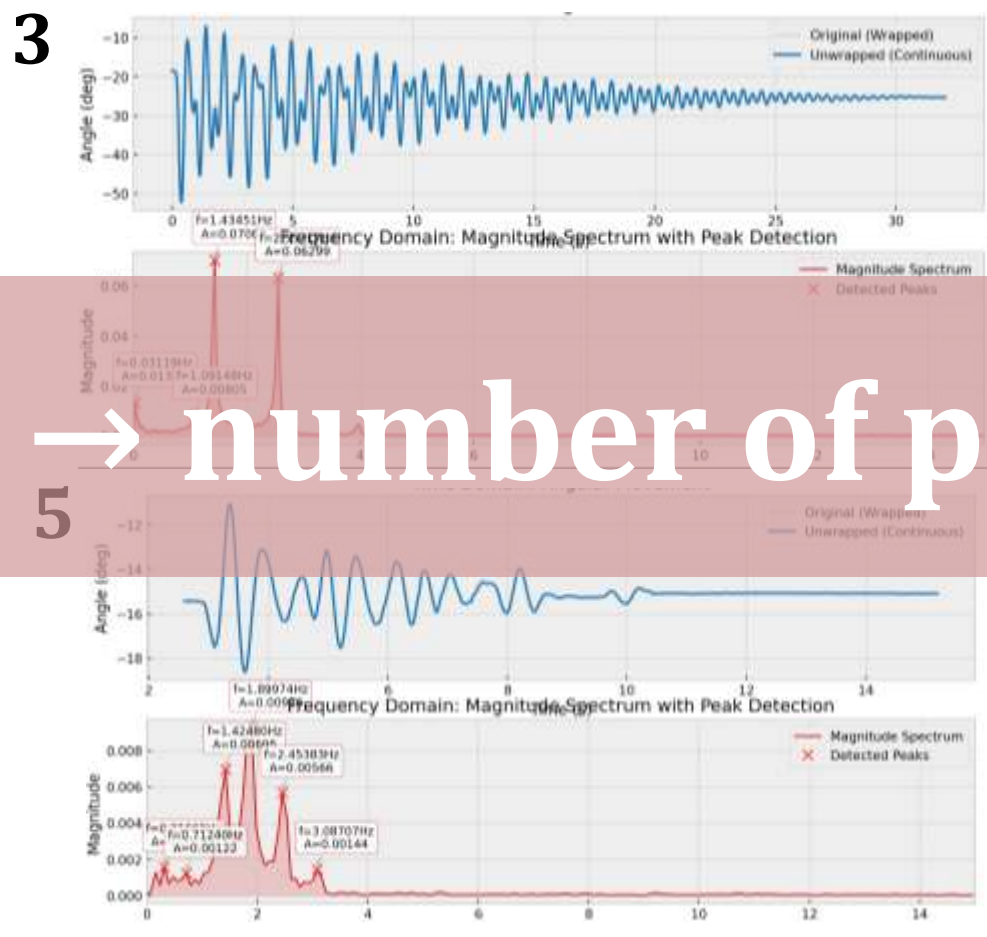
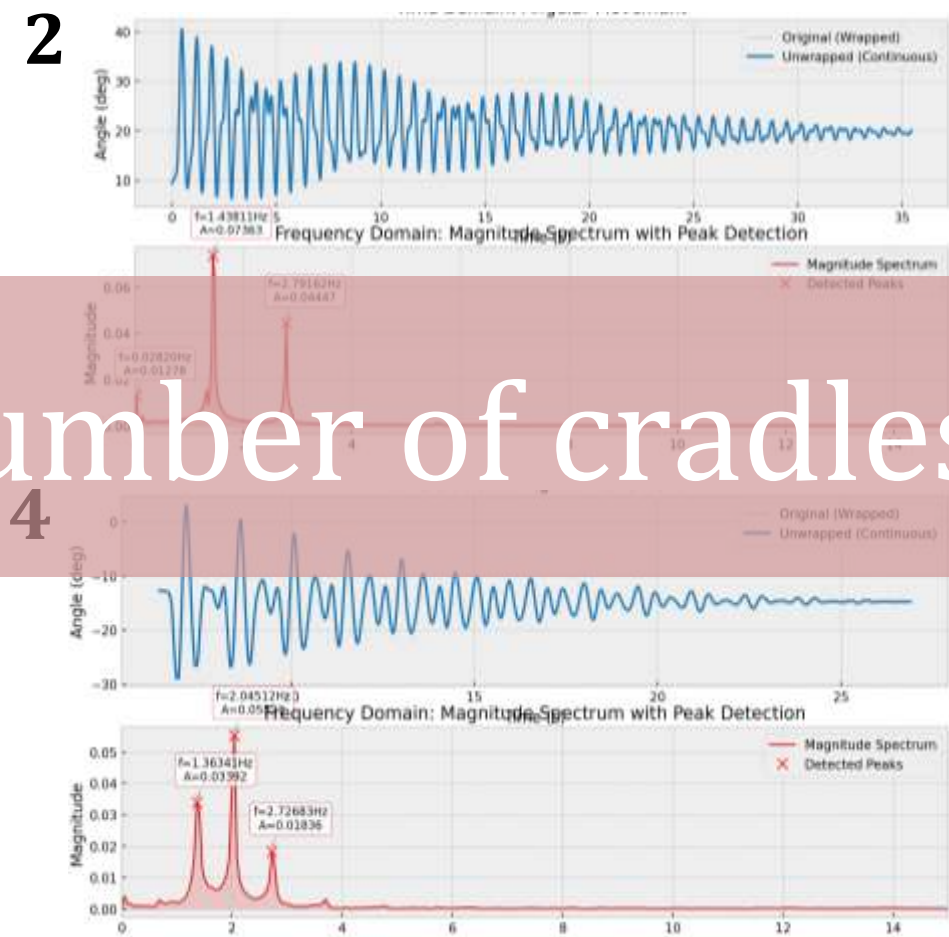
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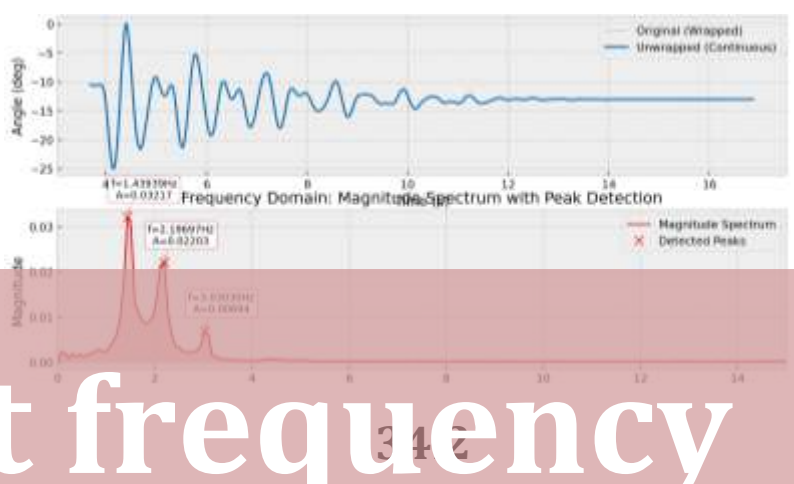
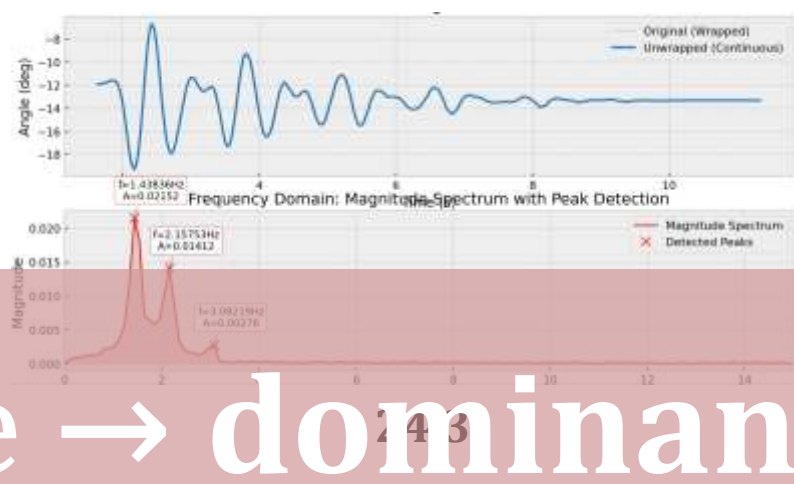
Number of cradles

Number of cradles → number of peaks

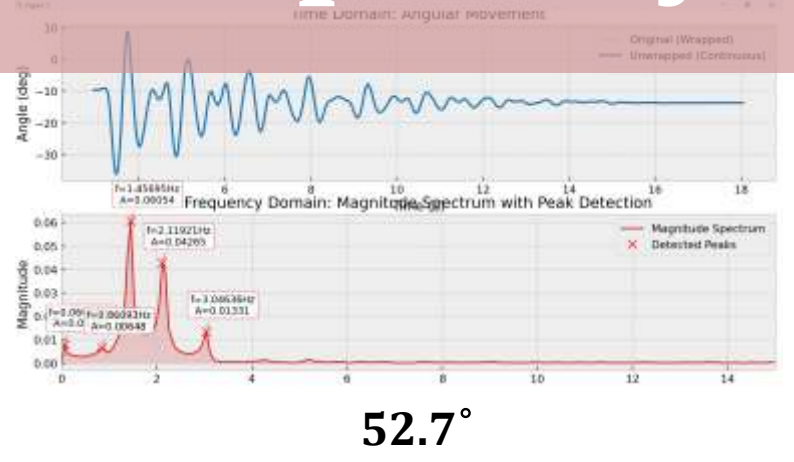
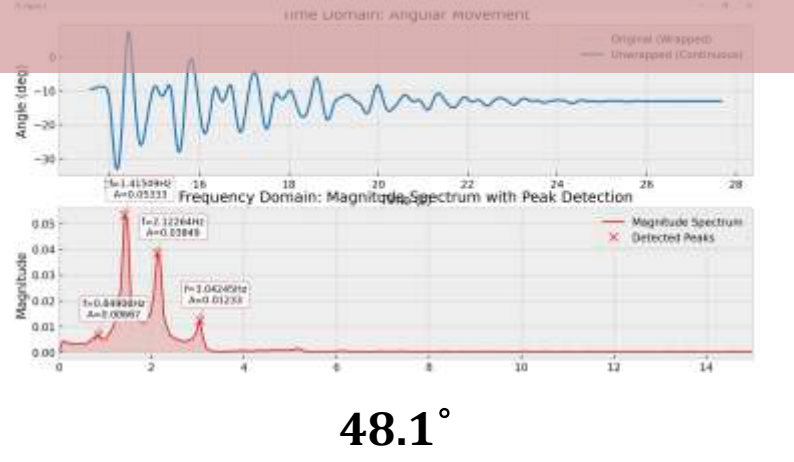
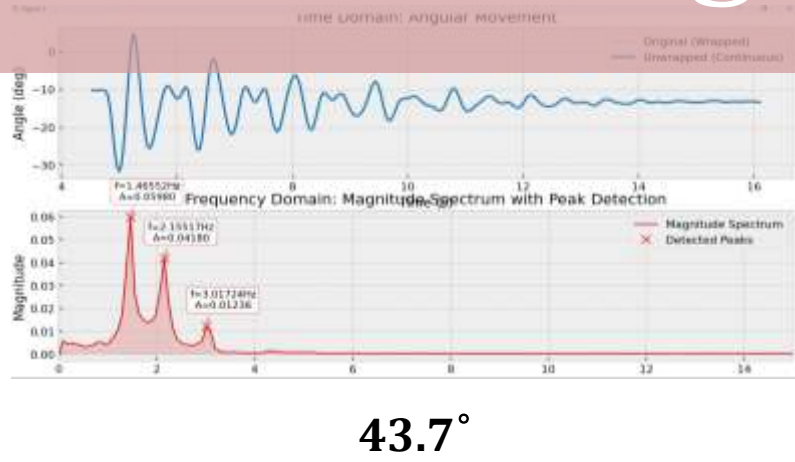


Preliminaries Results

Release angle



Release angle → dominant frequency



43.7°

48.1°

52.7°

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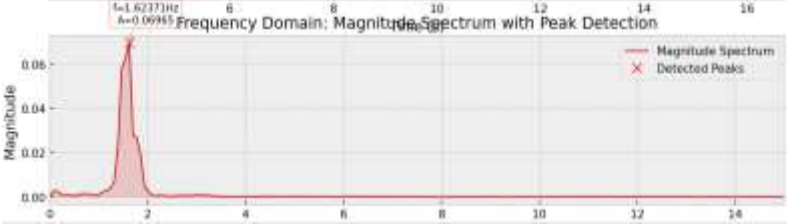
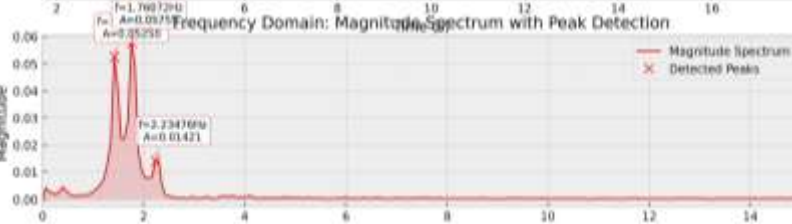
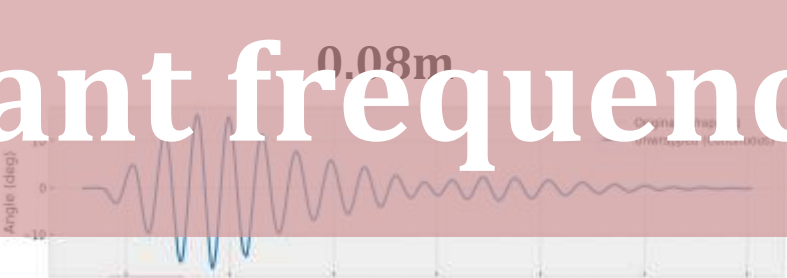
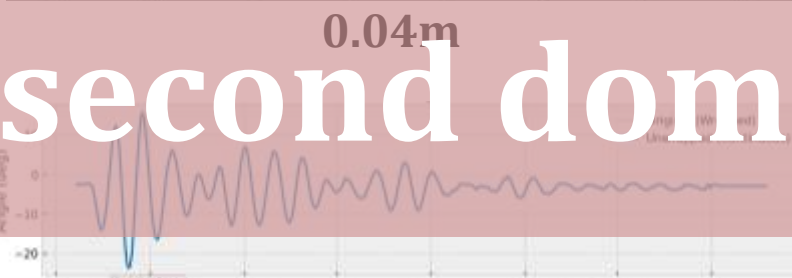
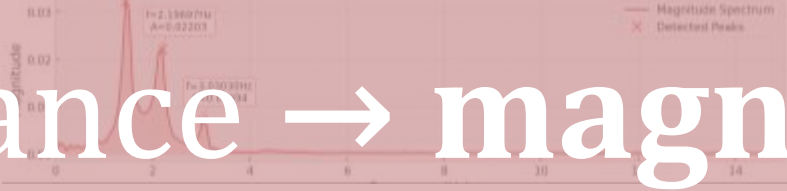
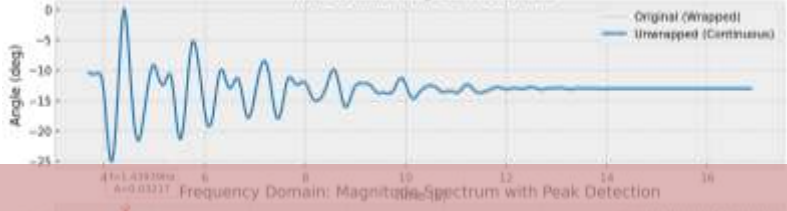
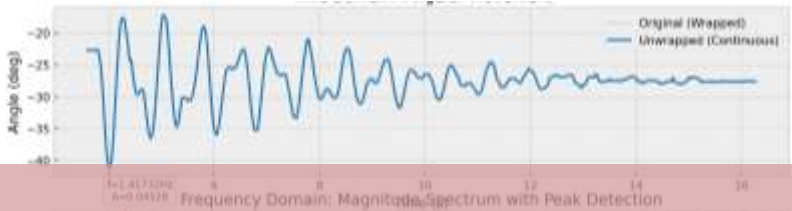
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Suspension point distance



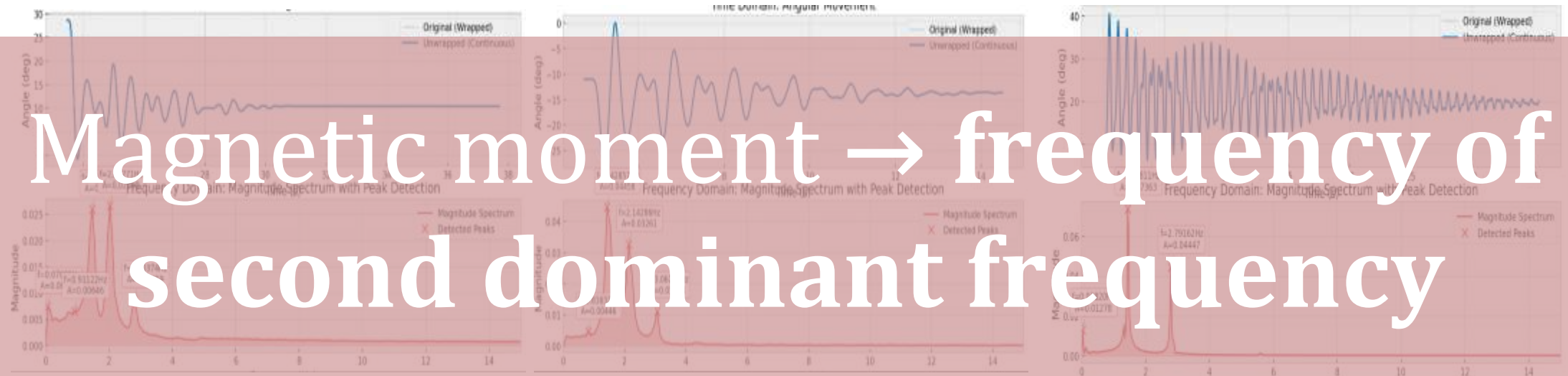
Suspension point distance → magnitude of second dominant frequency

Preliminaries Results

Magnetic moment

$$m = M \cdot V$$

Magnetic moment is determined by both geometric size and magnetization



4.08Am²

8.16Am²

16.31Am²

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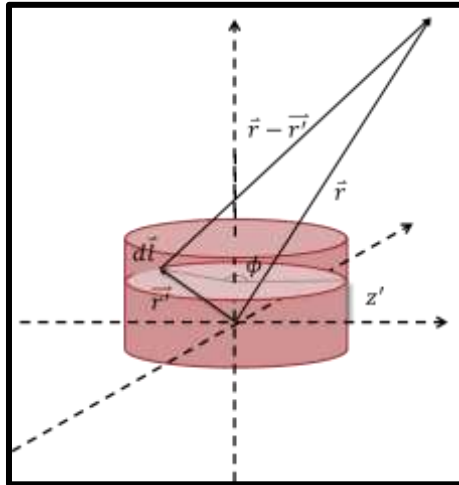
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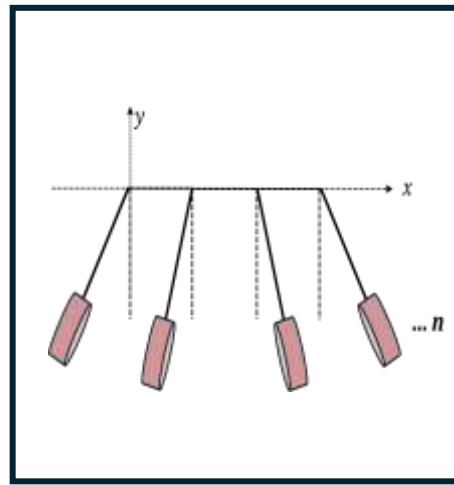
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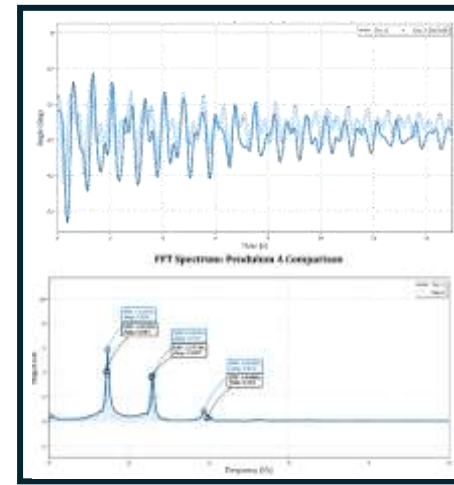
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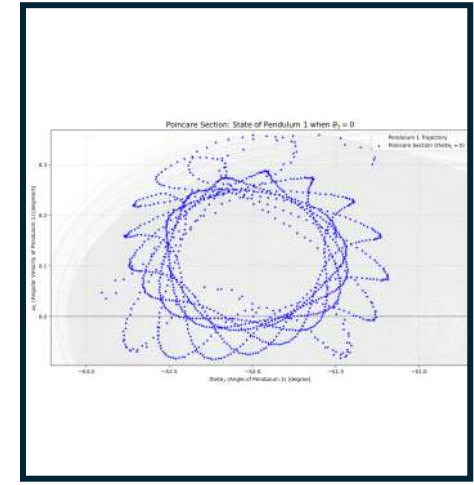
***Magnetic
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***Dynamical
Model***

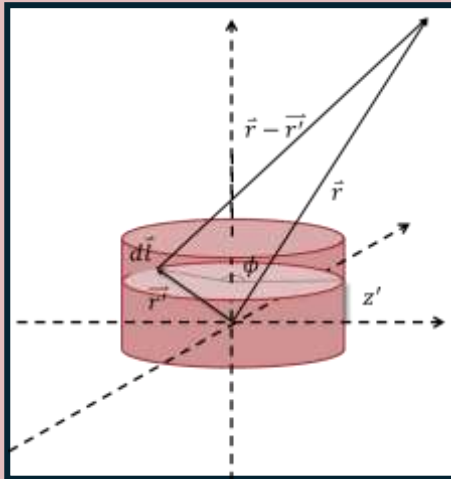


***Kinematic
Analysis***

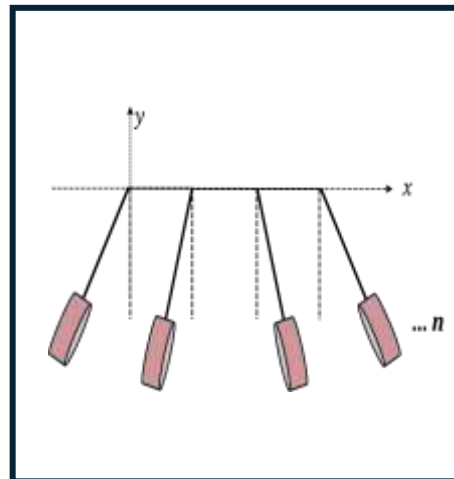


***Chaotic
Analysis***

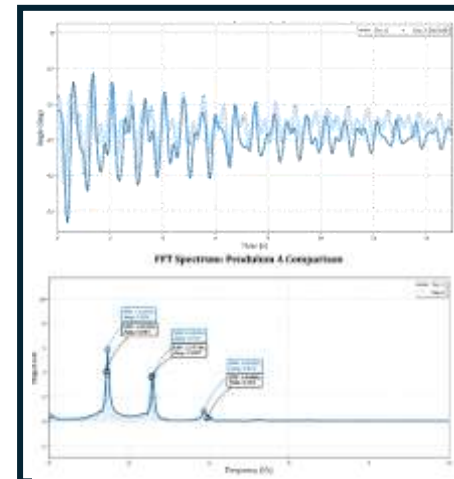
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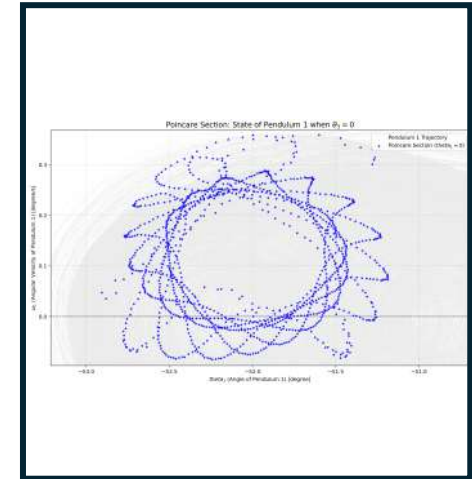
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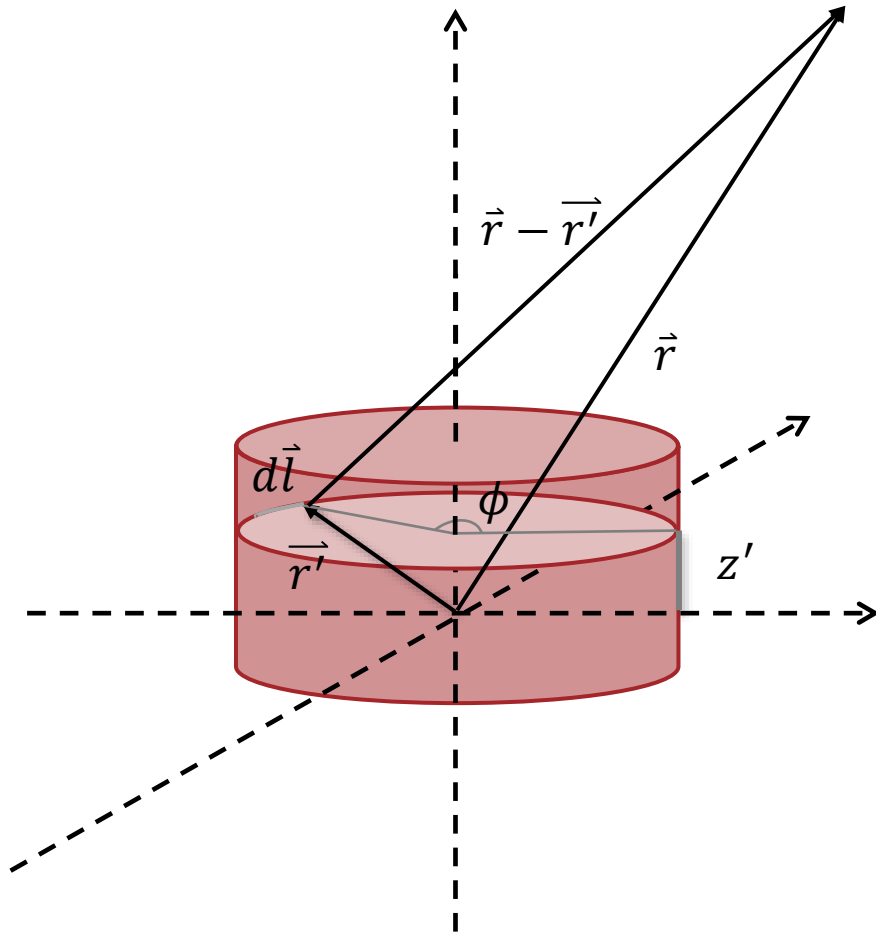


***Chaotic
Analysis***

Basic Assumptions

- ① The magnets are identical and **uniformly magnetized**
- ② The magnetic pendulum is subject to **linear damping**
- ③ The motion is restricted to a **two-dimensional plane**

Current Cylinder



Biot-Savart Law

$$\vec{r}' = (R \cos \phi, R \sin \phi, z')$$

$$d\vec{l}(\phi) = R d\phi (-\sin(\phi), \cos(\phi), 0)$$

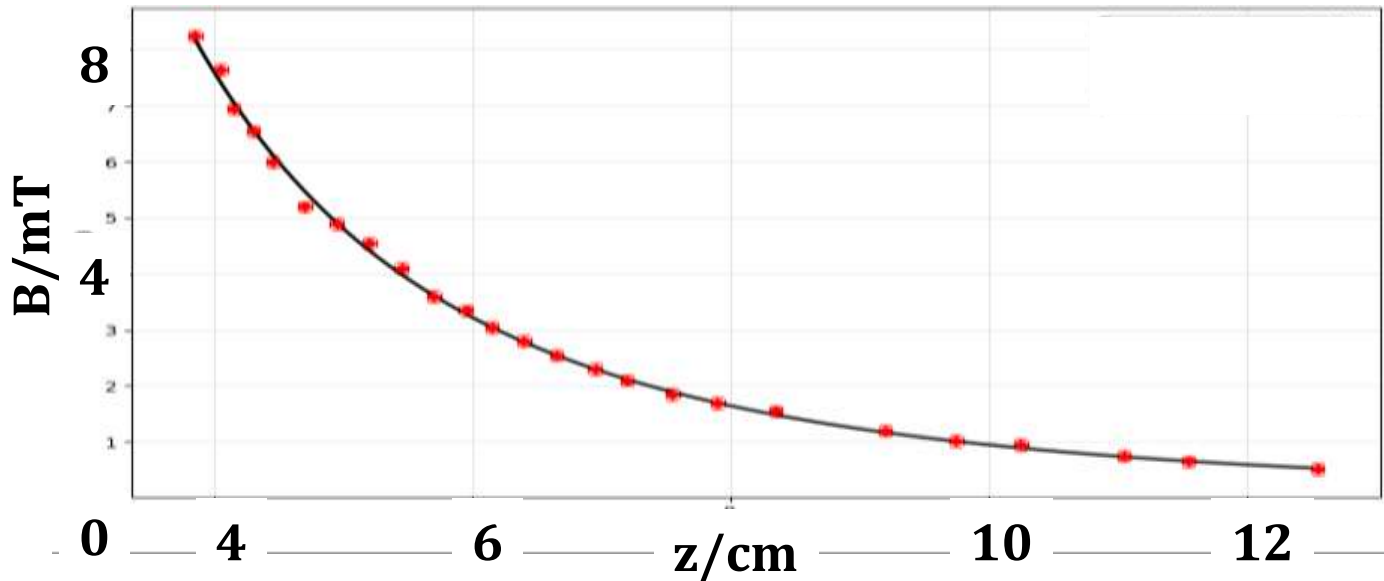
$$B(r) = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l}' \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

$$\vec{B}(\vec{r}) = \frac{\mu_0 I}{4\pi} \int_{z'=0}^L \int_{\phi=0}^{2\pi} \frac{d\vec{l}(\phi) \times [\vec{r} - (R \cos(\phi), R \sin(\phi), z')]}{|\vec{r} - (R \cos(\phi), R \sin(\phi), z')|^3} dz'$$

Magnet Field

Axial Magnetic Field

$$B_z(z) = \frac{\mu_0 M}{2} \left[\frac{z + \frac{H}{2}}{\sqrt{\left(z + \frac{H}{2}\right)^2 + R^2}} - \frac{z - \frac{H}{2}}{\sqrt{\left(z - \frac{H}{2}\right)^2 + R^2}} \right]$$



Magnetization

$$\mathbf{M} = (1.12 \pm 0.04) \cdot 10^6 \text{ A/m}$$

Equivalent Current

$$dI = K_b dz = M_{mag} dz$$

$$I_{eq} = \int_0^L dI = M_{mag} L$$

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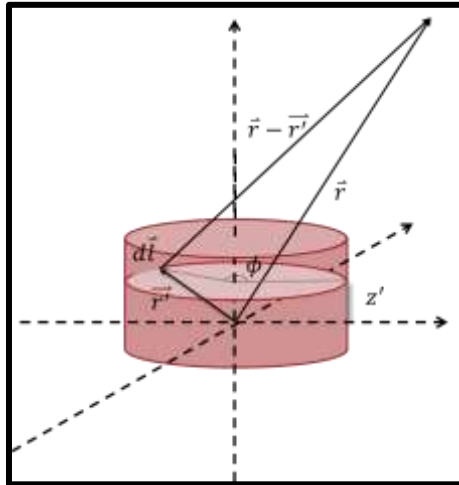
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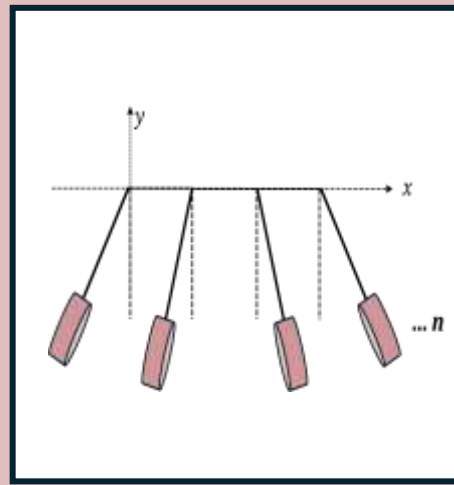
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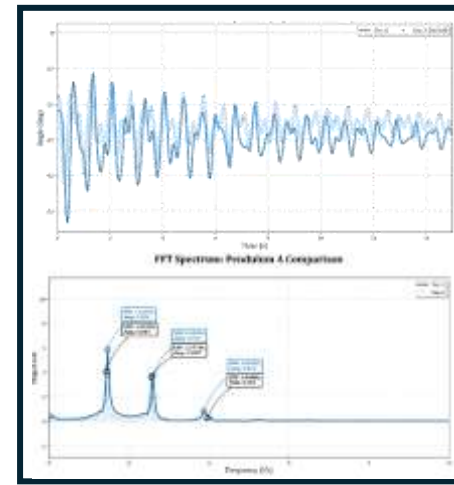
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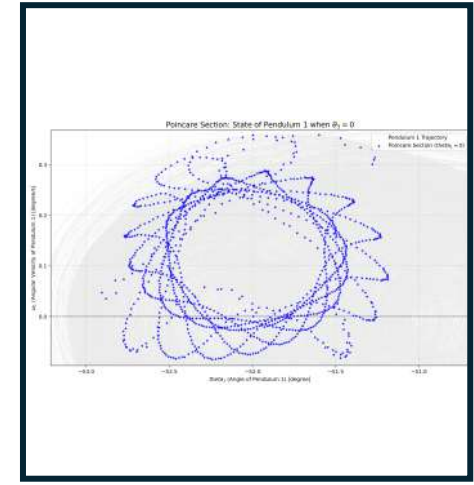
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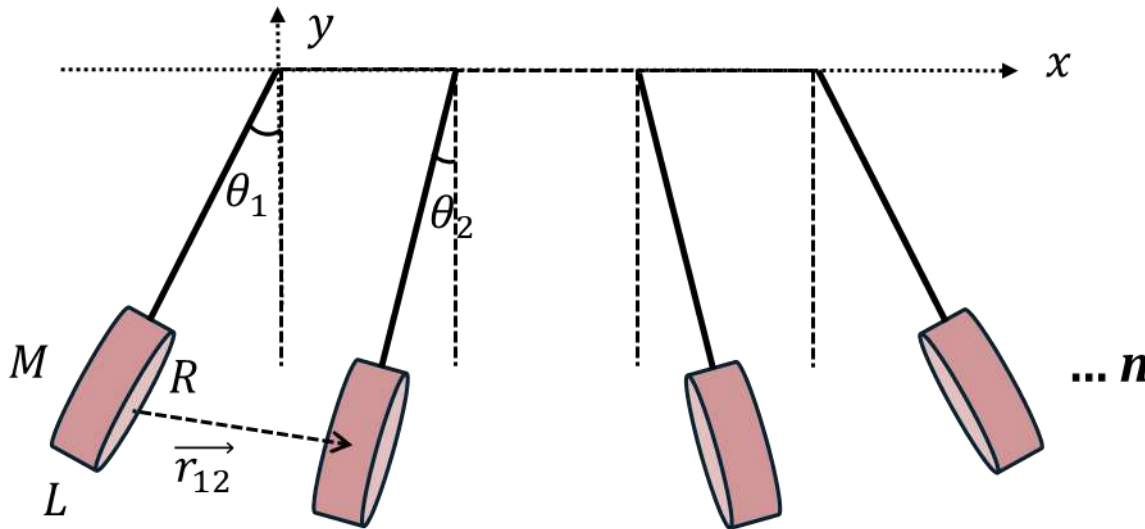
***Kinematic
Analysis***



***Chaotic
Analysis***

Kinetic Energy

n degrees of freedom $q_k = \{ \theta_1, \theta_2, \dots, \theta_n \}$



Moment of Inertia

$$I_{eq} = I_{cm} + Md^2 + \frac{1}{3} M_{rod} L^2$$

$$= M \left(\frac{1}{4} R^2 + \frac{1}{12} L^2 + (l + R)^2 \right) + \frac{1}{3} M_{rod} L^2$$

Kinetic Energy

$$T = \frac{1}{2} I_{eq} \dot{\theta}^2 = \frac{1}{2} I_{eq} \sum_{i=1}^n \dot{\theta}_i^2$$

Position Vector

$$\mathbf{p}_i = ((i-1)s + l \sin \theta_i, -l \cos \theta_i, 0)$$

$$r_{ij} = |\Delta \mathbf{p}_{ij}| = \sqrt{\left((j-i)s + l(\sin \theta_j - \sin \theta_i) \right)^2 + \left(l(\cos \theta_i - \cos \theta_j) \right)^2}$$

Potential Energy

Gravitational Potential Energy

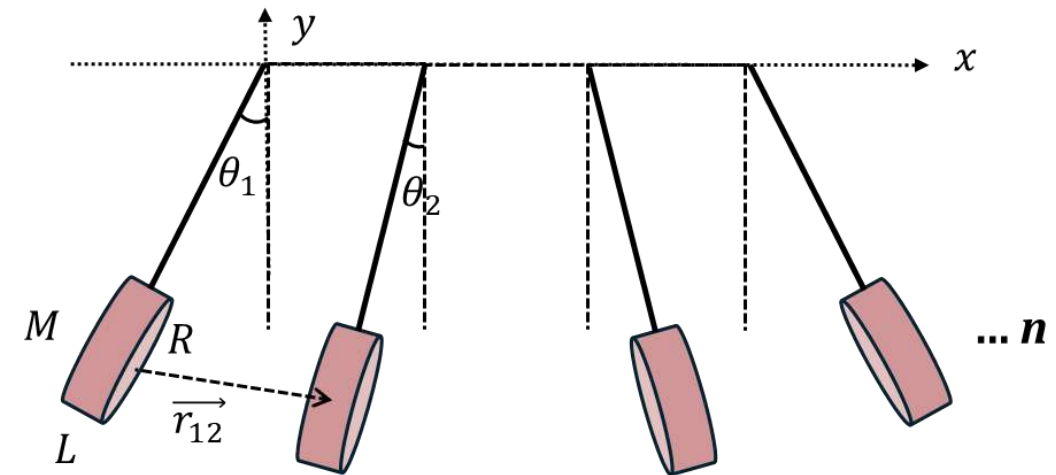
$$V_{grav} = \sum_{i=1}^n Mg(l + R)(1 - \cos \theta_i)$$

Magnetic Potential Energy

$$V_{mag,ij} = - \iiint_{V_j} \overrightarrow{M_{mag,j}} \cdot \overrightarrow{B_i}(\vec{r}) dV$$

Total Potential Energy

$$V = V_{grav} + \sum_{j < i} V_{mag,ij}$$



Lagrangian Equation

Lagrangian

$$L = T - V = \underbrace{\frac{1}{2} I_{eq} \sum_{i=1}^n \dot{\theta}_i^2}_{\text{Kinetic Energy}} - \underbrace{\sum_{i=1}^n M g (l + R) (1 - \cos \theta_i)}_{\text{Gravitational Potential Energy}} - \underbrace{\sum_{j < i} V_{mag,ij}}_{\text{Magnetic Potential Energy}}$$

Rayleigh Dissipation Function

$$\mathcal{F} = \frac{1}{2} \sum \gamma \dot{\theta}_i^2$$

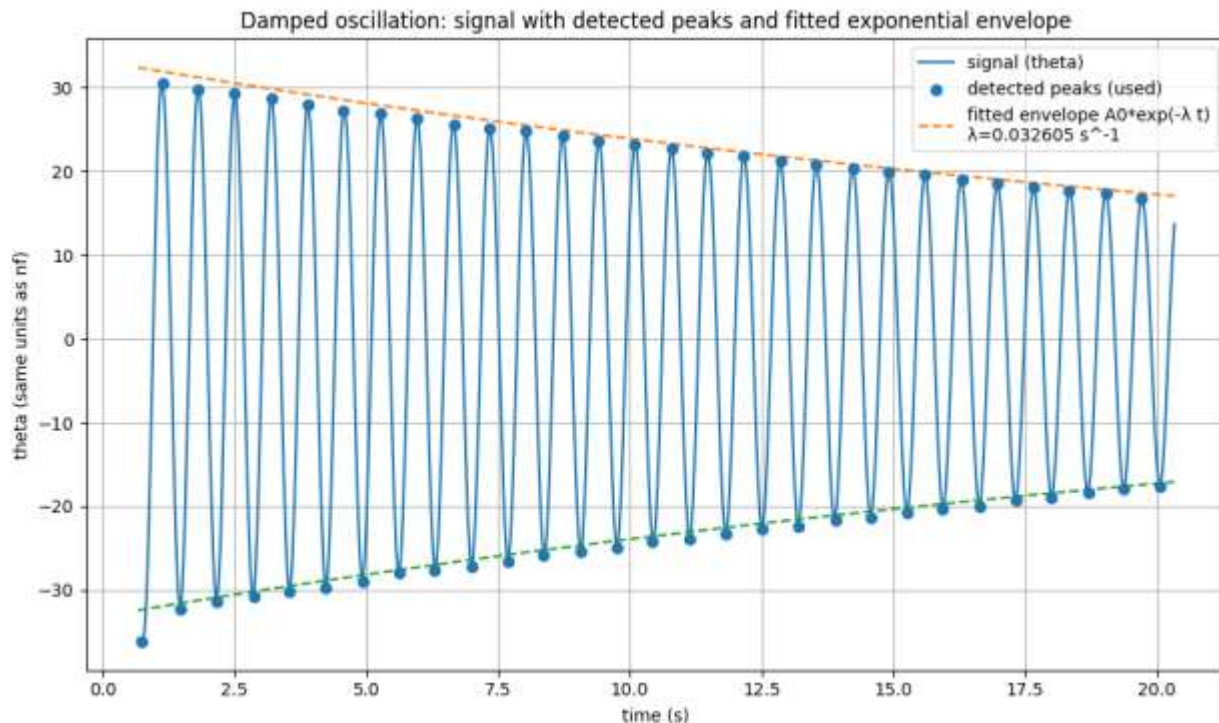
$$\frac{\partial \mathcal{F}}{\partial \dot{\theta}_k} = \gamma \dot{\theta}_k$$

Equations of Motion

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_k} \right) - \frac{\partial L}{\partial \theta_k} + \frac{\partial \mathcal{F}}{\partial \dot{\theta}_k} = 0 \quad \left\{ \begin{array}{l} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) - \frac{\partial L}{\partial \theta_1} + \frac{\partial \mathcal{F}}{\partial \dot{\theta}_1} = 0 \\ I_{eq} \ddot{\theta}_n + \frac{\partial V(\ddot{\theta})}{\partial \theta_n} + \gamma \dot{\theta}_n = 0 \end{array} \right.$$

Damping Coefficient

Calculated through experiment by letting a single cradle swing freely



Renolds Number

$$Re \approx 556$$

Damping Coefficient

$$\theta = \theta_0 \cdot e^{\lambda \cdot t}$$

$$\lambda = 0.0326 \pm 0.0009 \text{ s}^{-1}$$

$$\gamma = 2I_{eq}\lambda = (4.33 \pm 0.12) \times 10^{-5} \text{ kg} \cdot \text{m}^2/\text{s}$$

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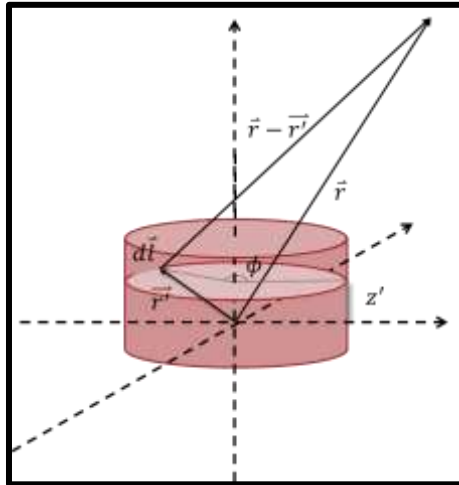
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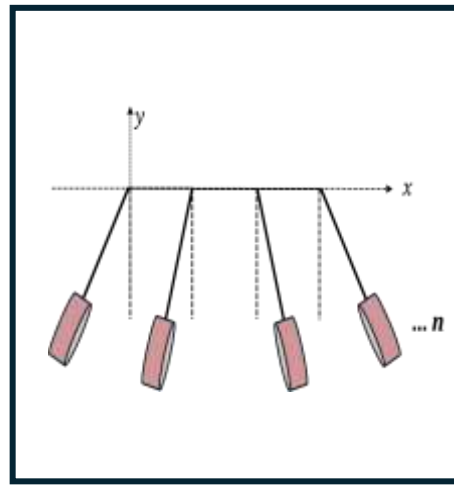
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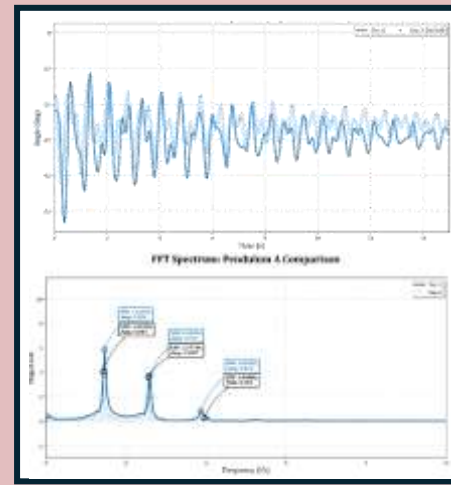
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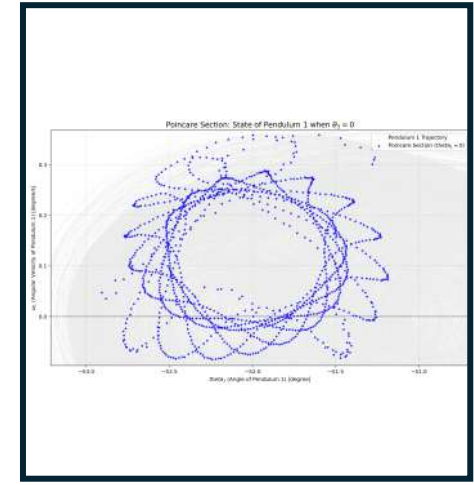
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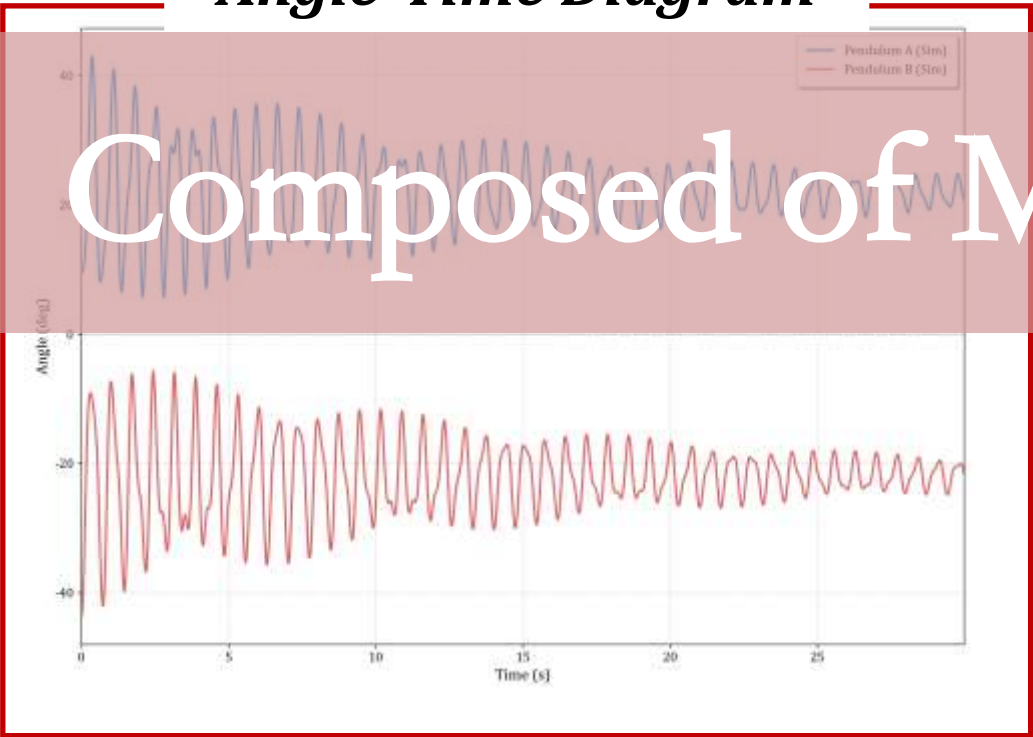


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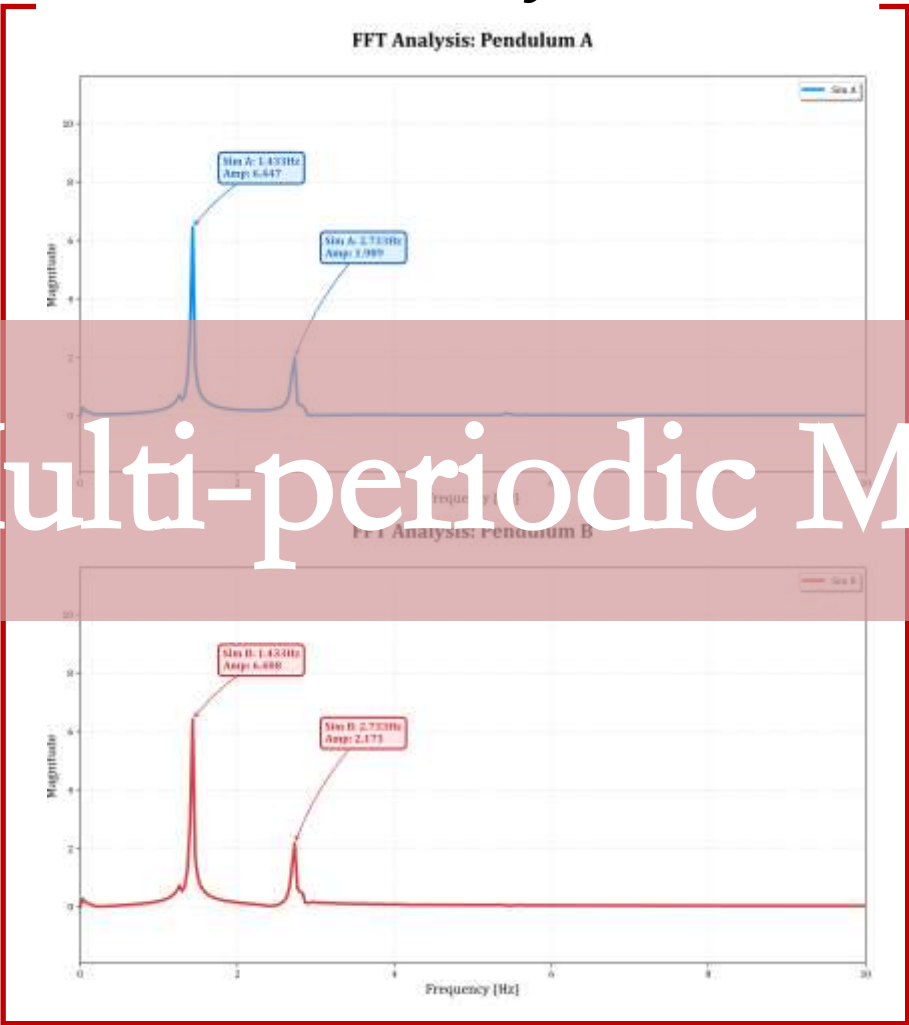
Motion Analysis

Use ***RK4*** to solve the differential equations of motion

Angle-Time Diagram



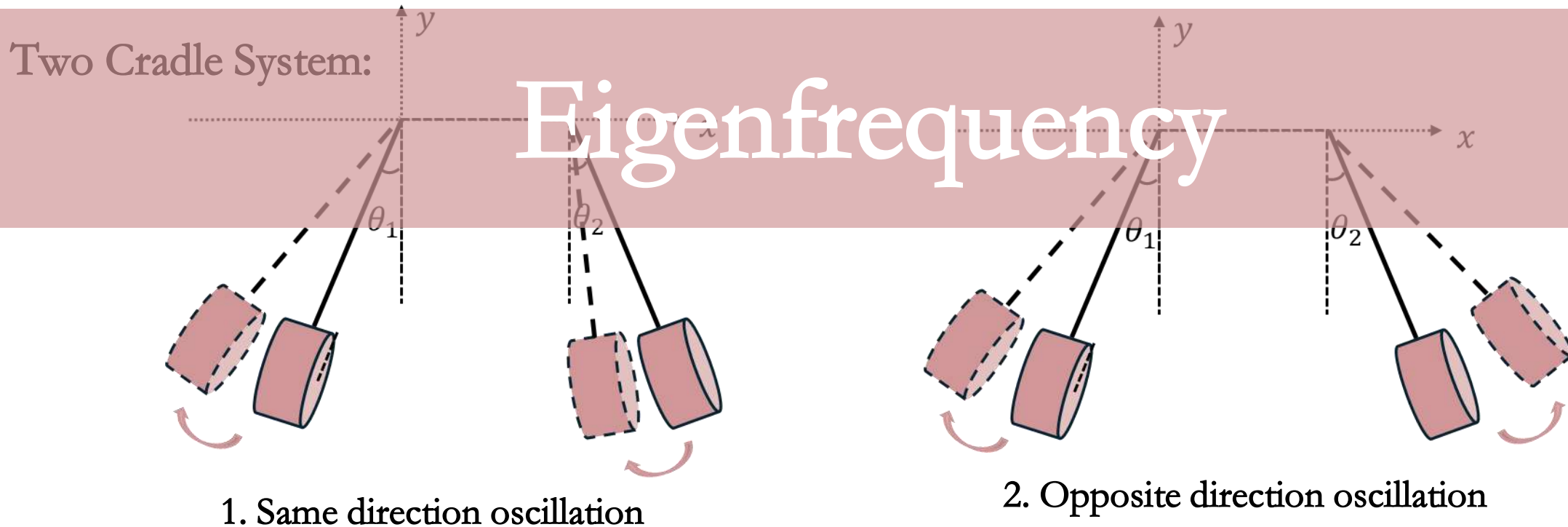
FFT Analysis



Two dominant frequency: 1.433, 2.733
Motions is composed of multi-periodic motion

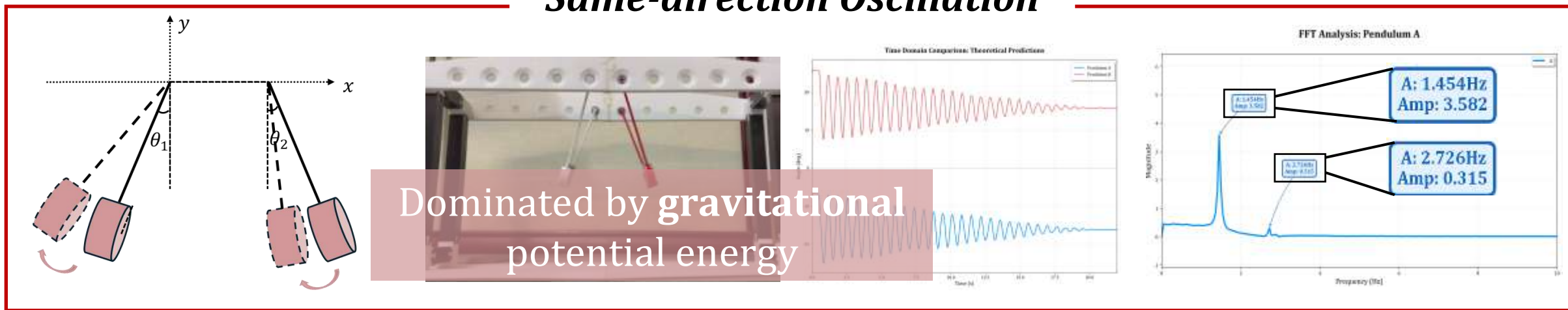
Normal Modes

- A coupled system possesses a set of natural frequencies (eigenfrequencies). Any initial state can be expressed as a linear superposition of these normal modes.
- **Number of eigenfrequency is equal to that of the DOF of the system**

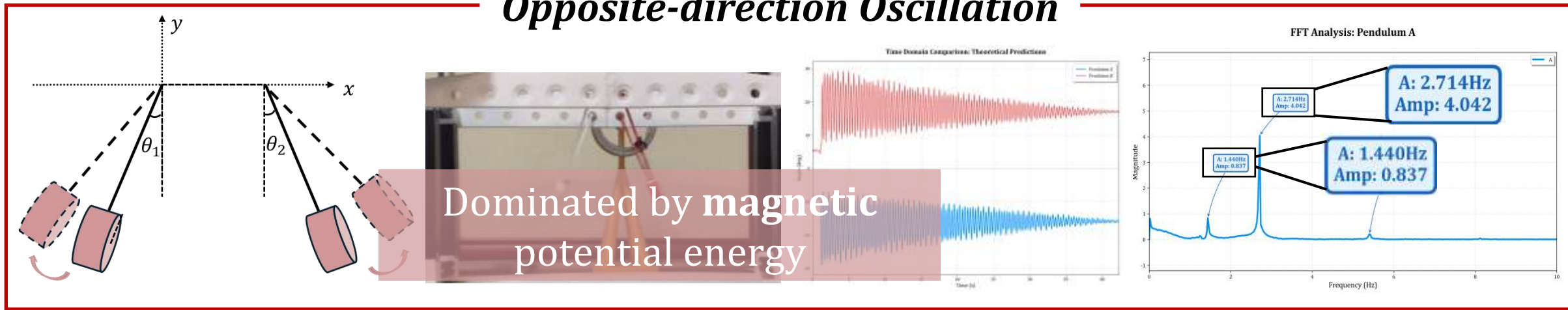


Mode Analysis

Same-direction Oscillation

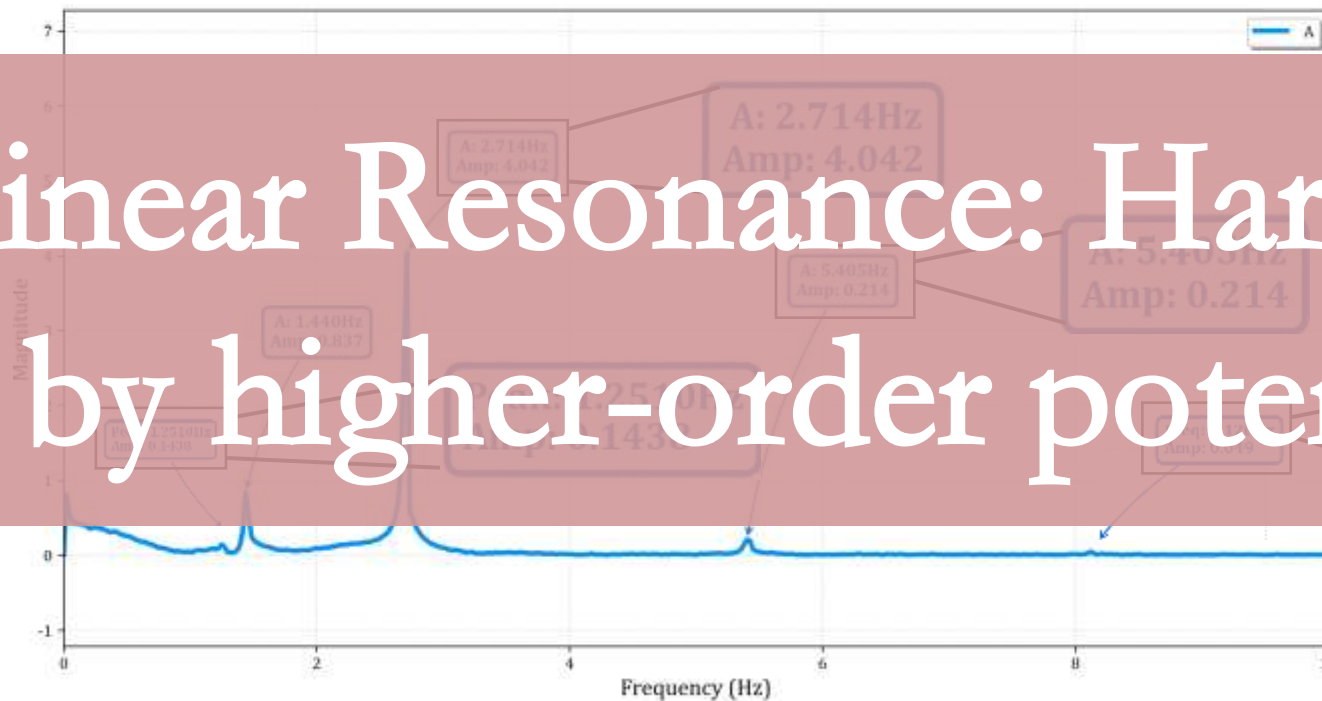


Opposite-direction Oscillation



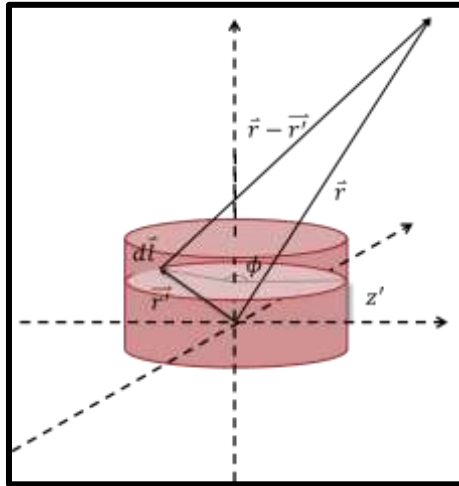
Nonlinear Behavior

FFT Analysis: Pendulum A

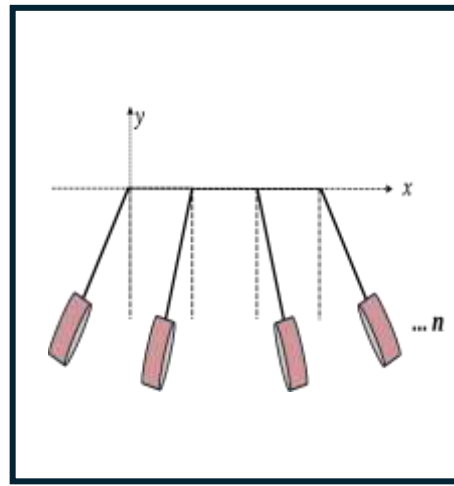


Nonlinear Resonance: Harmonics induced by higher-order potential terms

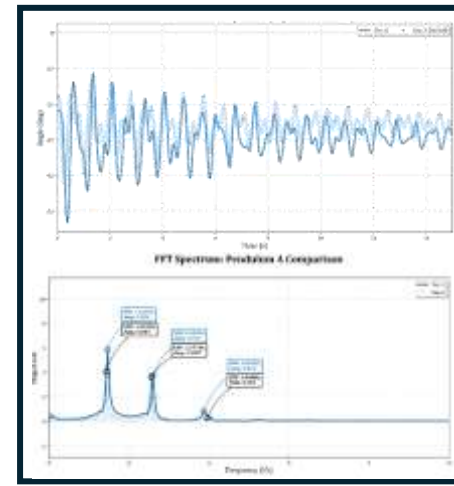
Theoretical analysis



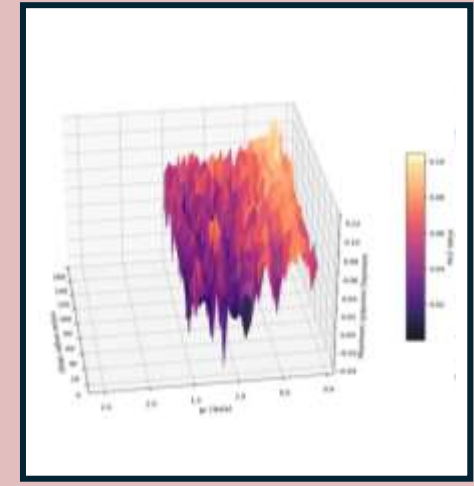
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Maximum Lyapunov Exponent

Maximum Lyapunov Exponent

$$|\delta(t)| \approx |\delta(0)| e^{\lambda t}$$

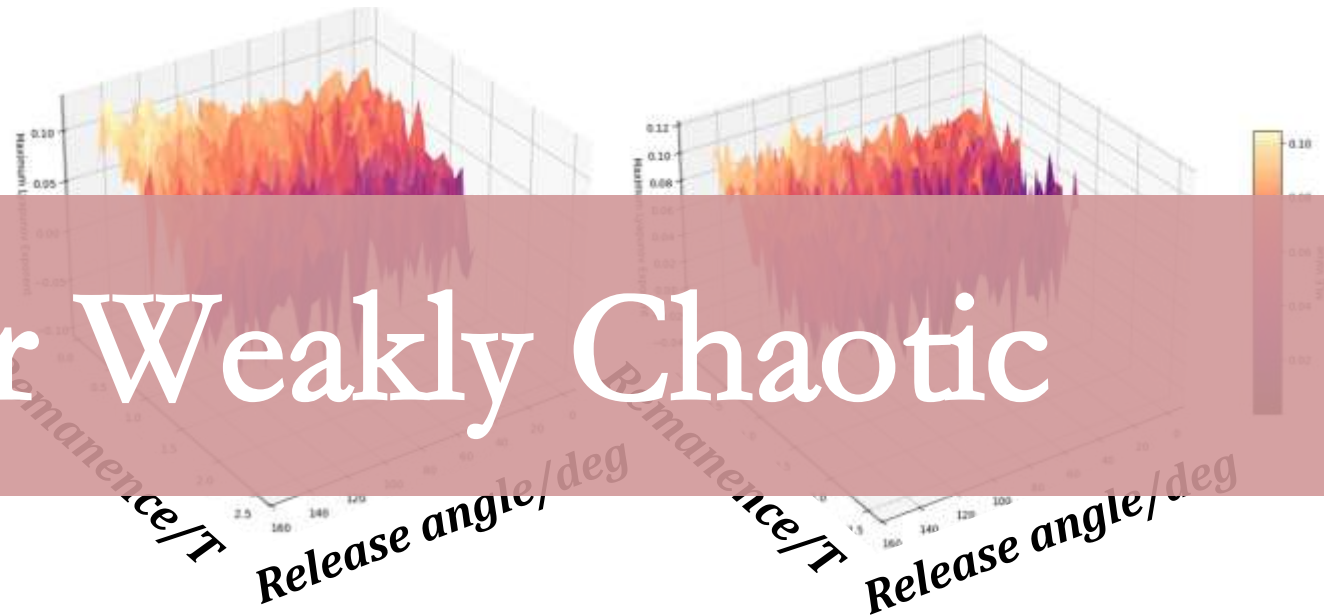
$$\lambda = \lim_{t \rightarrow \infty} \lim_{\delta(0) \rightarrow 0} \frac{1}{t} \ln \frac{|\delta(t)|}{|\delta(0)|}$$

Non-chaotic or Weakly Chaotic

If $\lambda > 0$, trajectories diverge exponentially, indicating chaotic behavior.

$N=5$

$N=8$



low magnetization and large initial release angles, $MLE > 0$



Number of magnets has no significant impact

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Poincaré Section

Poincaré section of cradle 1 at $\theta_3 = 0$ (5-cradle system)

Low magnetization

$I=1500$ A

Low magnetization

$I=5500$ A

Low magnetization

$I=5500$ A

$\omega_1 / \text{deg s}^{-1}$

Non-chaotic or Weakly Chaotic

$\omega_1 / \text{deg s}^{-1}$

θ_1 / deg

θ_1 / deg

θ_1 / deg

Weak response

Quasi-periodic loop

Nonlinear Resonance

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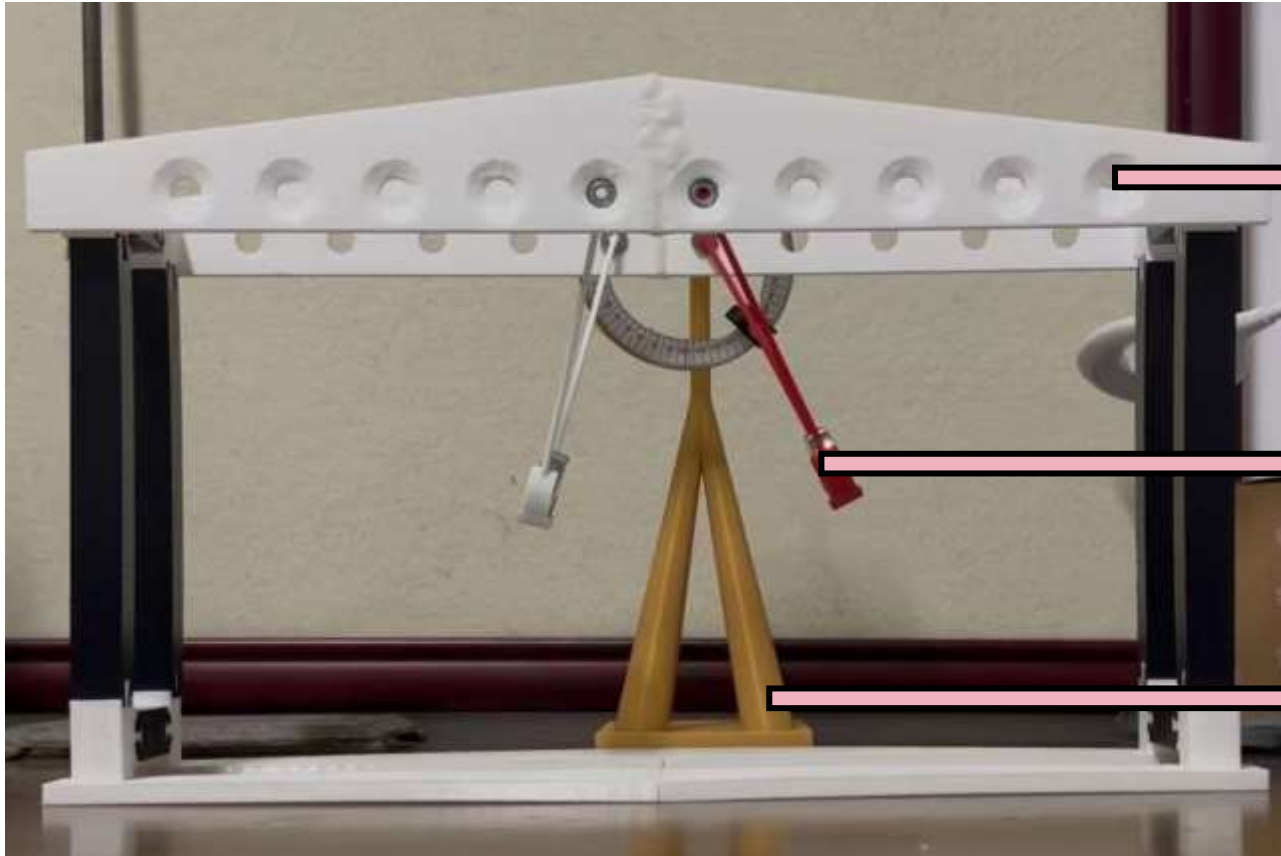
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Experimental Setup



3D-printed PETG Mount
Non-ferromagnetic Material



3D-printed cradle holder



protractor



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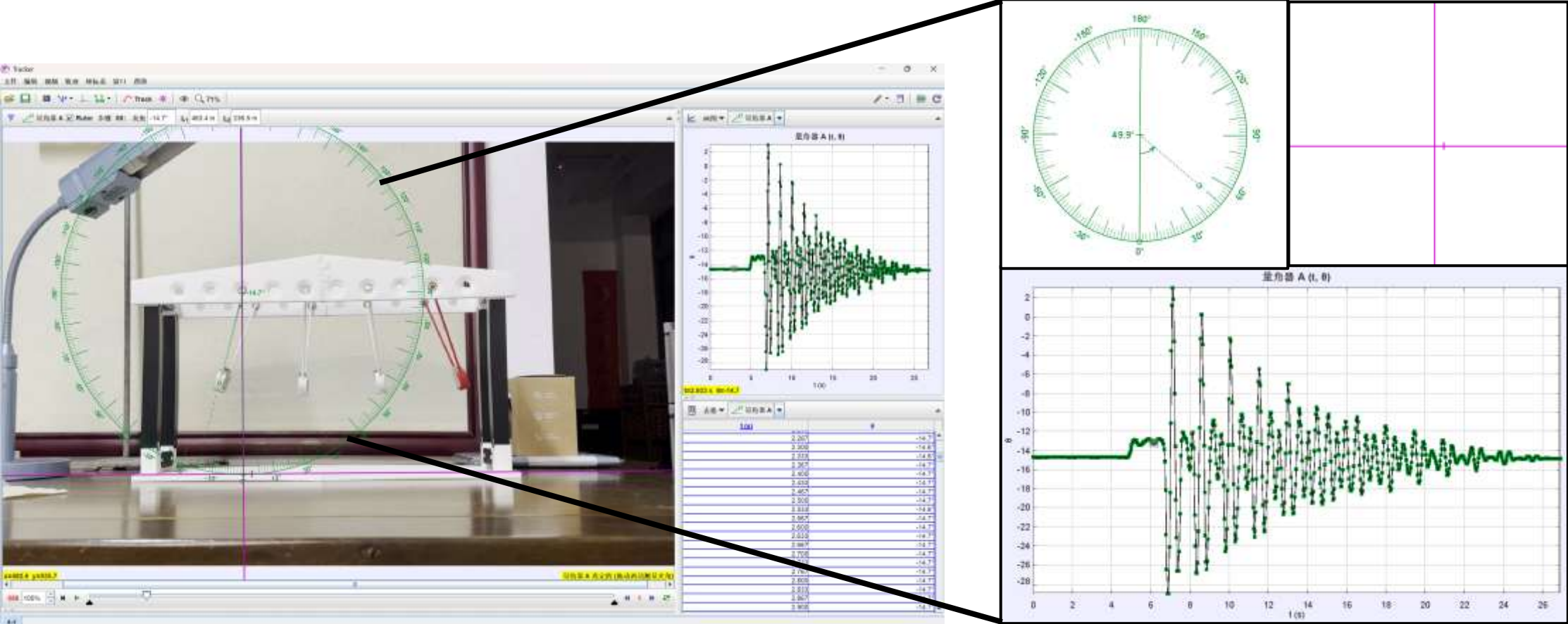
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Video Analysis --- Tracker



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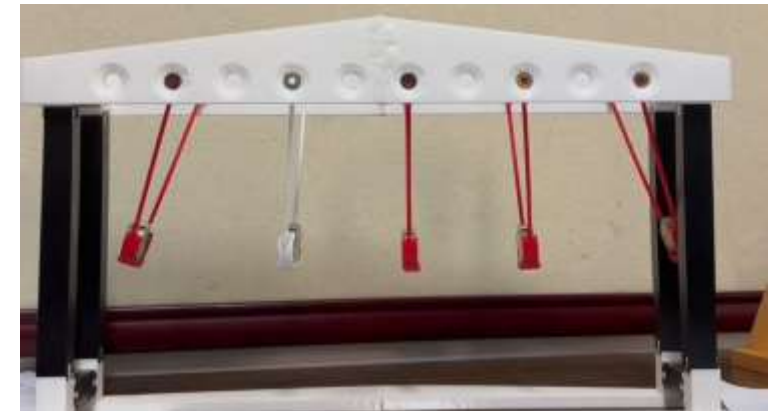
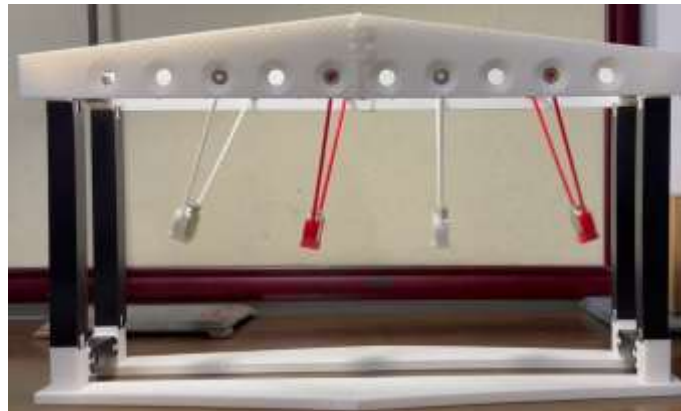
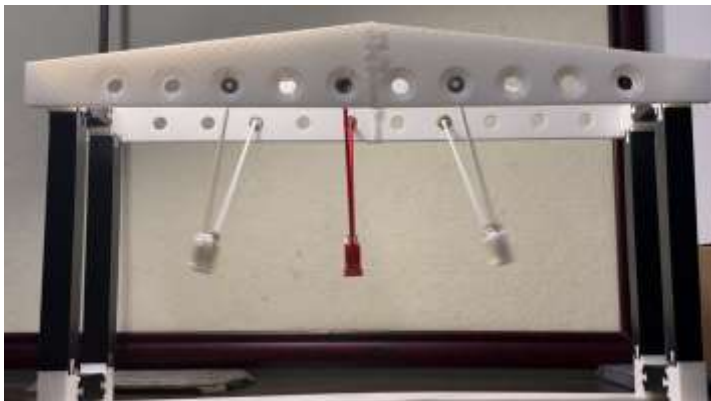
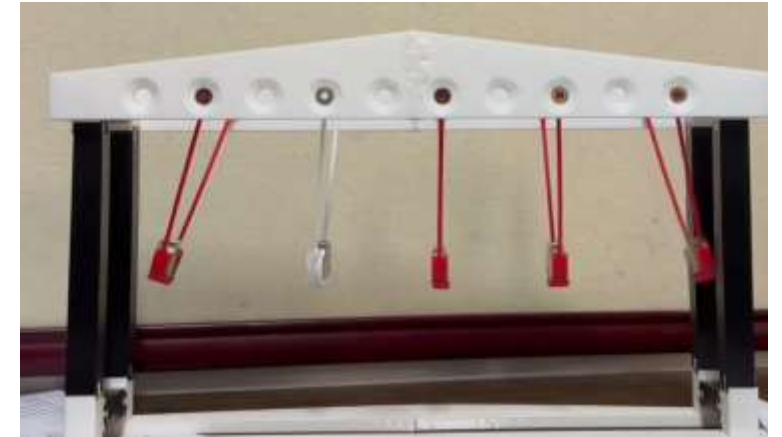
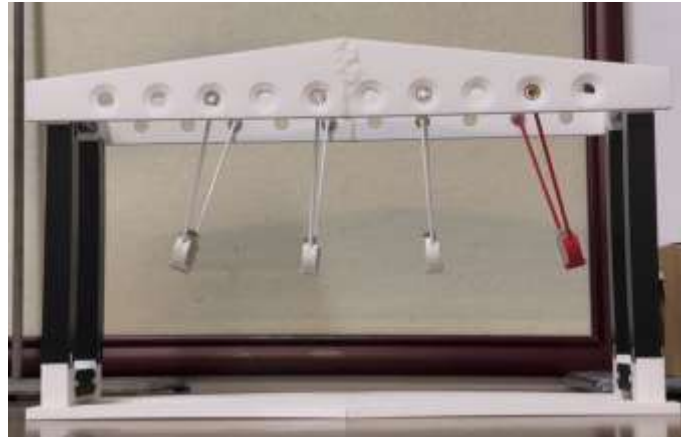
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Conduct multiple magnetic cradle experiments



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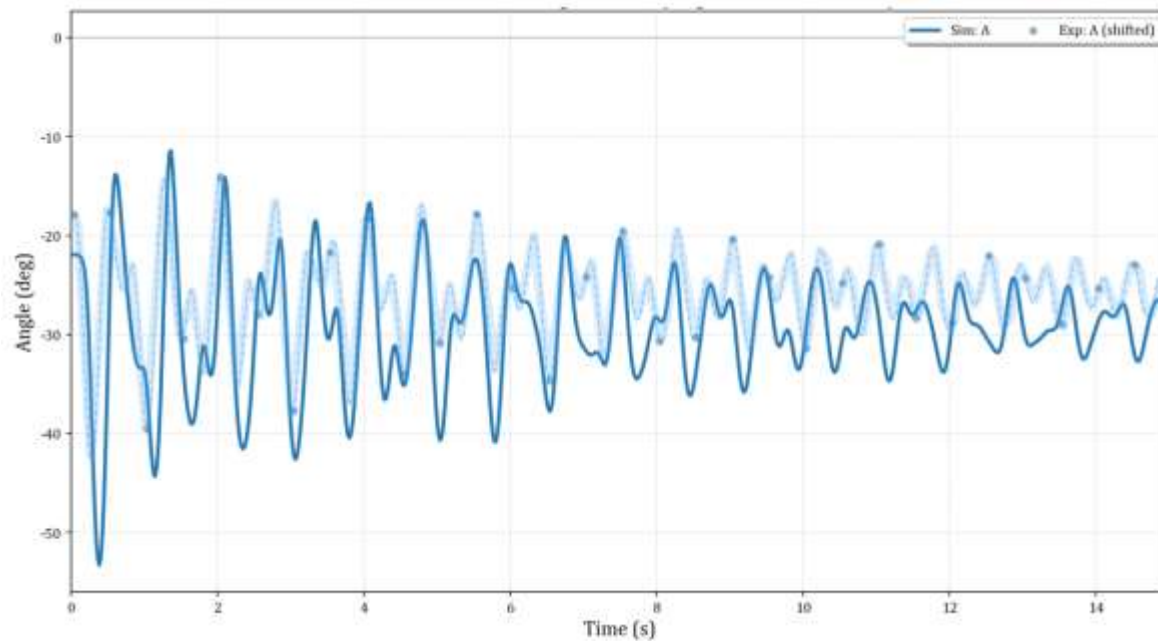
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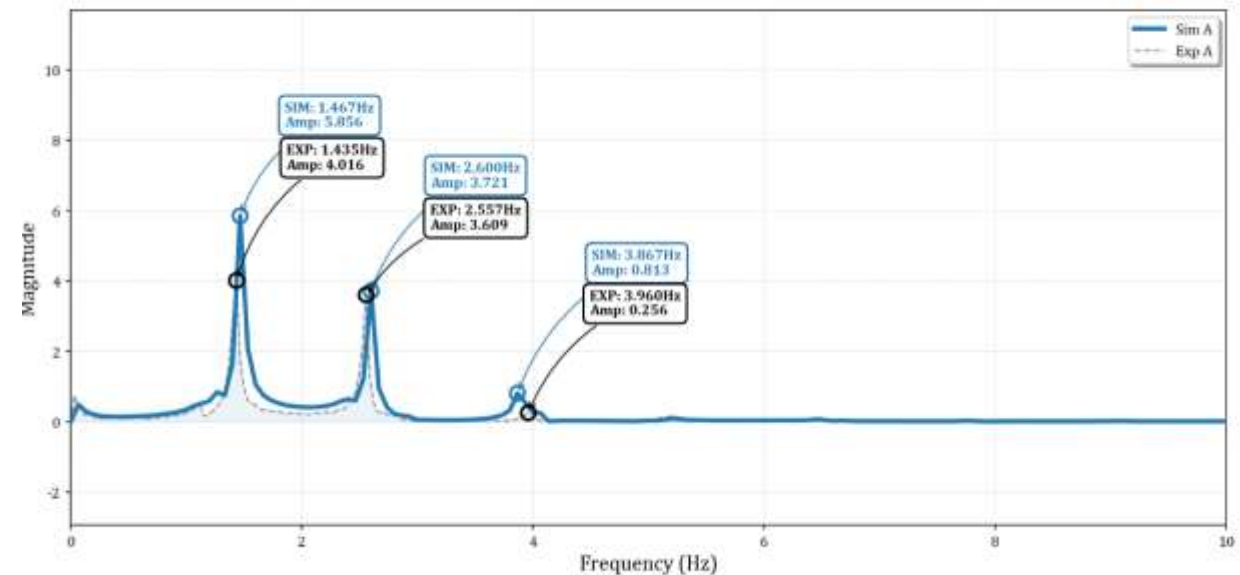
Experiment VS Simulation - Dynamics

3 cradles

Compare experimental data with theory → well-aligned



FFT Spectrum: Pendulum A Comparison



Relative error 2.09%

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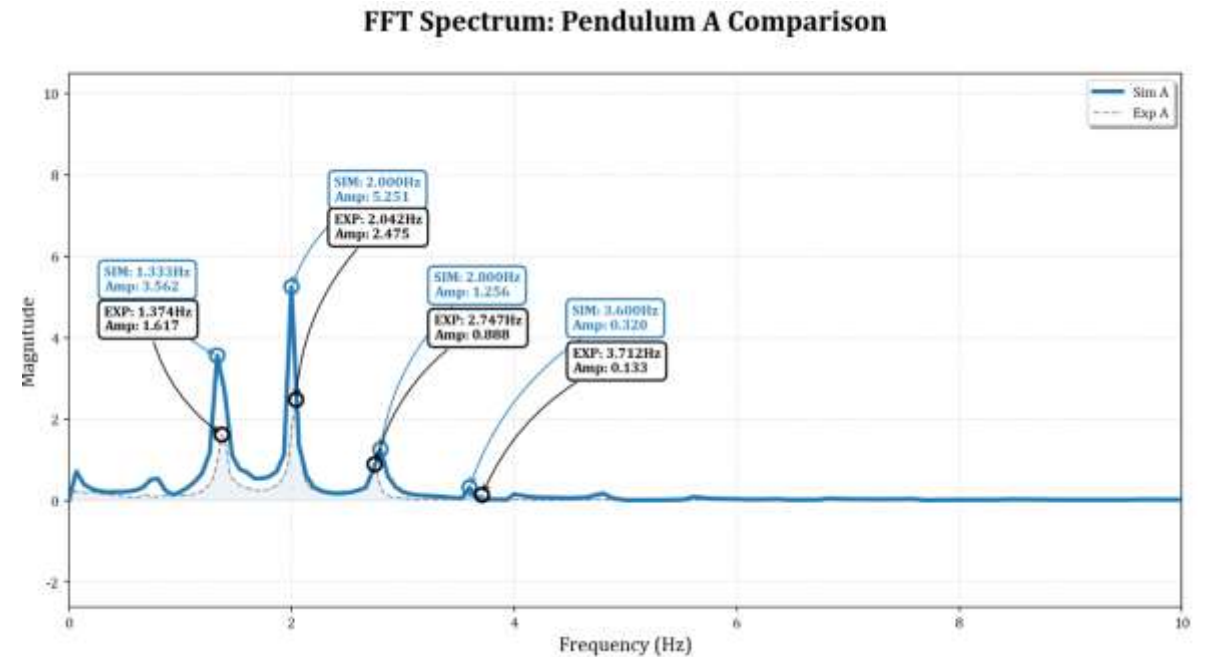
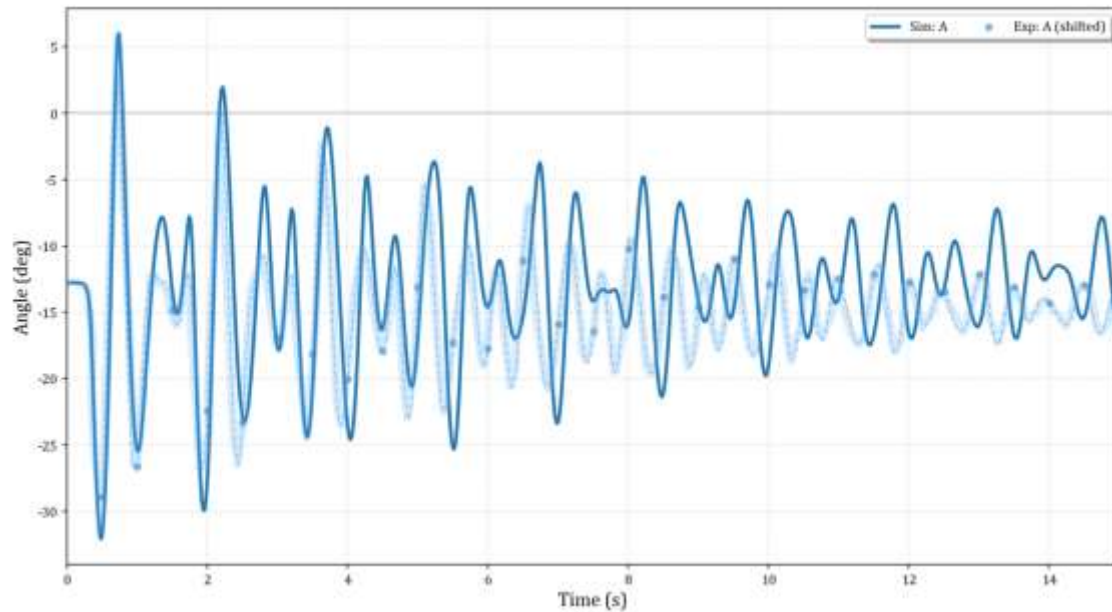
Experiment

Conclusion

Experiment VS Simulation - Dynamics

4 cradles

Compare experimental data with theory → well-aligned



Relative error 2.50%

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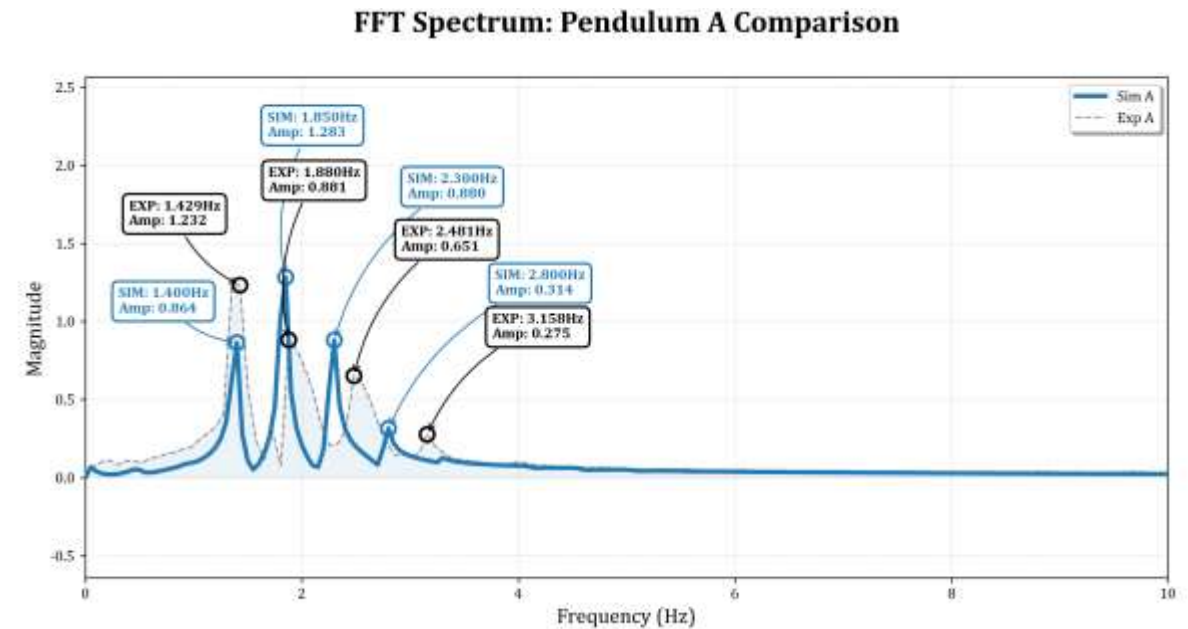
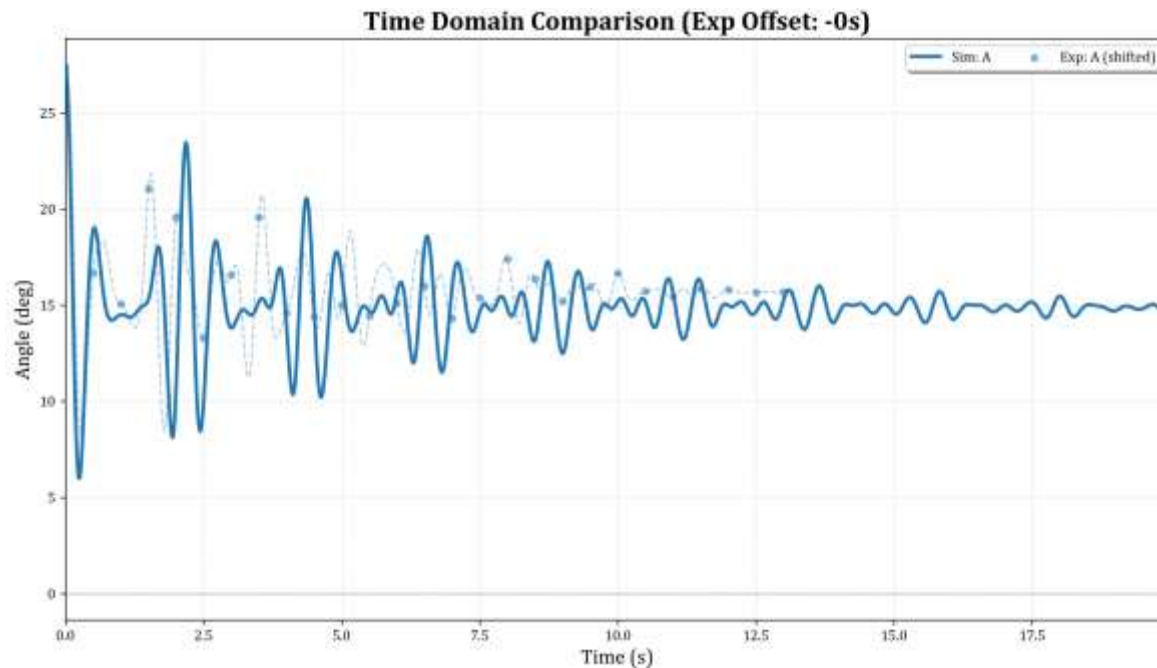
Experiment

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Experiment VS Simulation - Dynamics

5 cradles

Compare experimental data with theory → well-aligned



Relative error 6.09%

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Experiment VS Simulation - Eigenfrequency

Parameters

Cradle 3
 $\theta_0=30.3^\circ$

Eigenfrequency

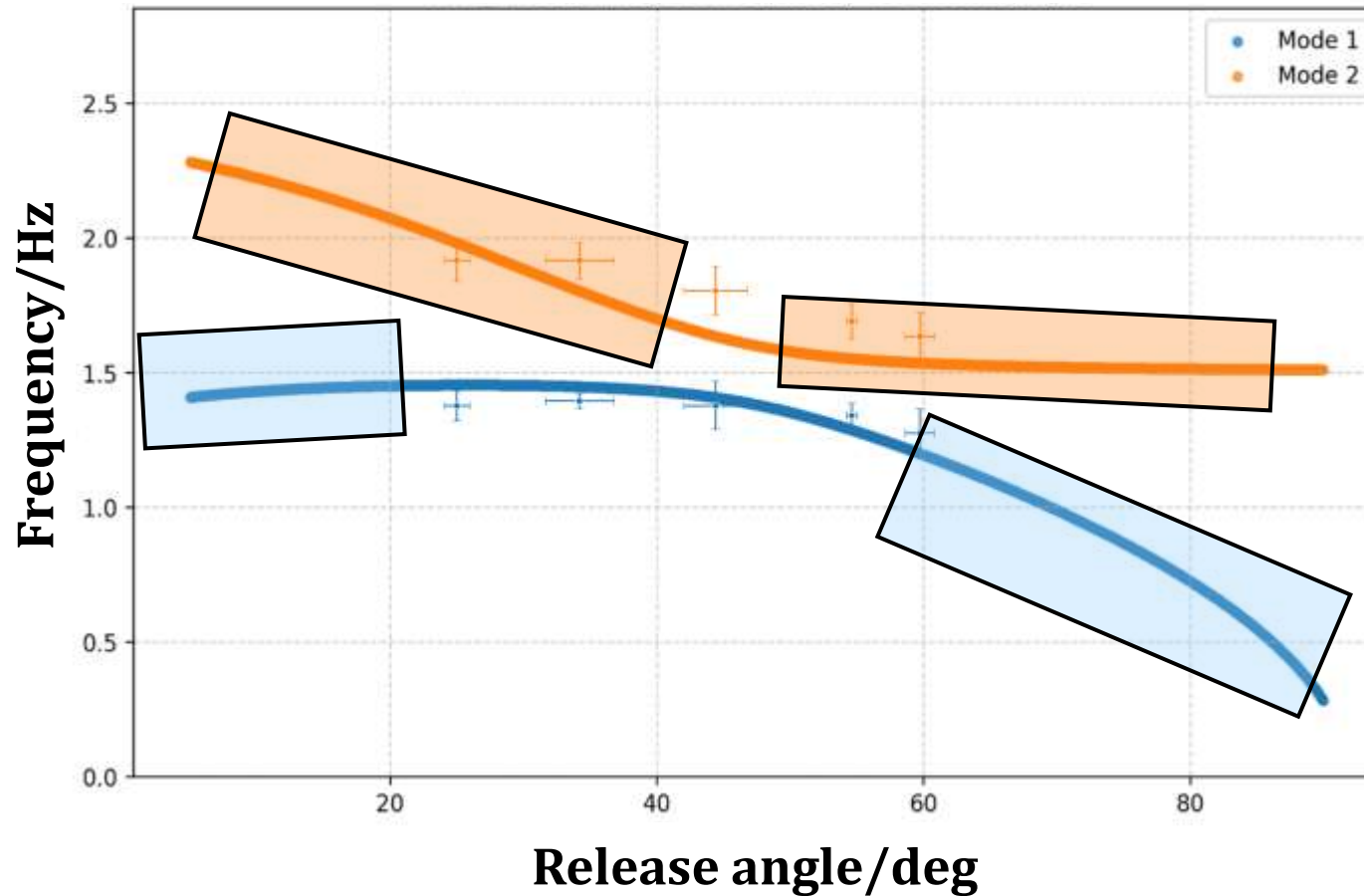
Mode	Simulation	Experiment	Absolute error	Relative error
1	1.425	1.432	0.007	-0.5%
2	2.492	2.546	0.05	-2%
3	3.896	3.952	0.06	-1%

well-aligned

Parameter Analysis

Release angle

Releasing angle's influence on 1st 2nd frequencies



Parameters

$N=3$, $s=0.08\text{m}$, $l=0.089\text{m}$,

$R=0.015\text{m}$, $L=0.01\text{m}$

Magnetic restoring stiffness always exists

Magnetic coupling effect weakens

Nonlinearity at large angles

Effect of high-order magnetic terms in total potential energy

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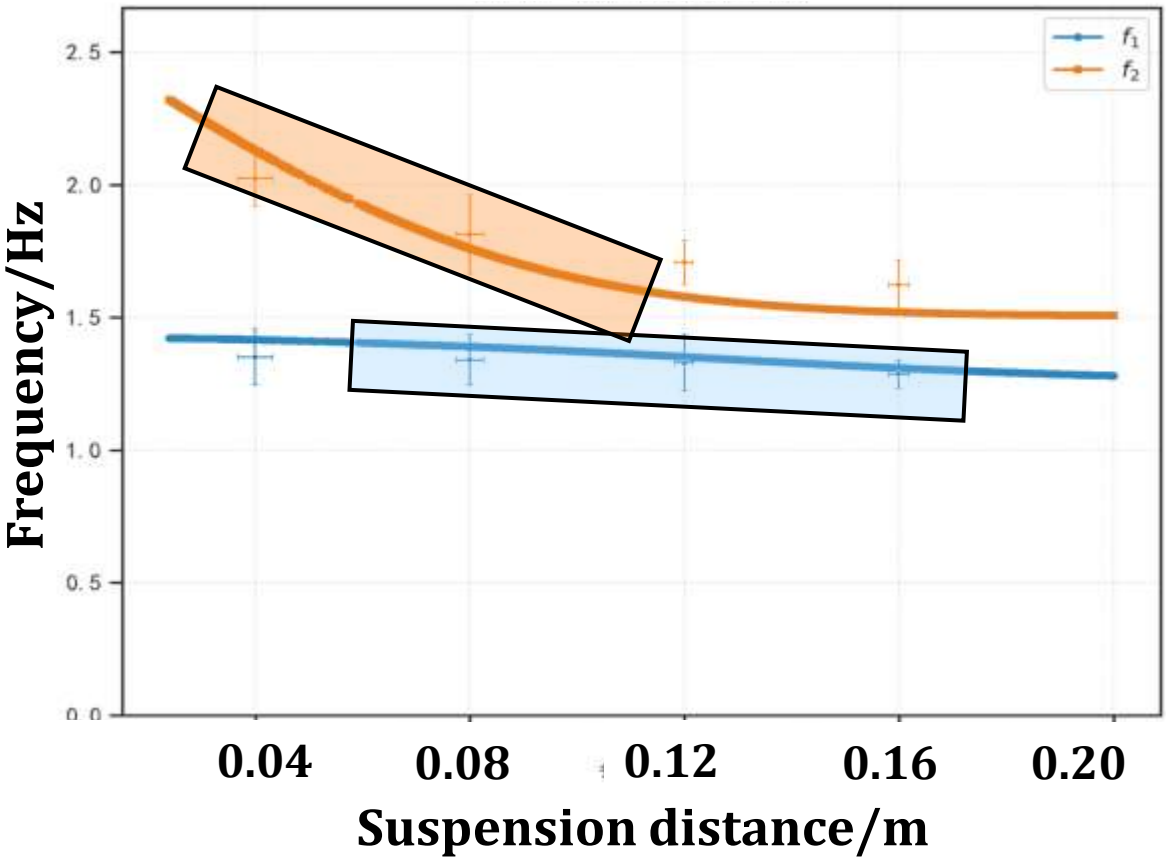
Parameter Analysis

Suspension distance

Parameters

N=3, $\theta_0=50$, $l=0.089\text{m}$,
R=0.015m, L=0.01m

Suspension Distance’s influence on 1st 2nd frequencies



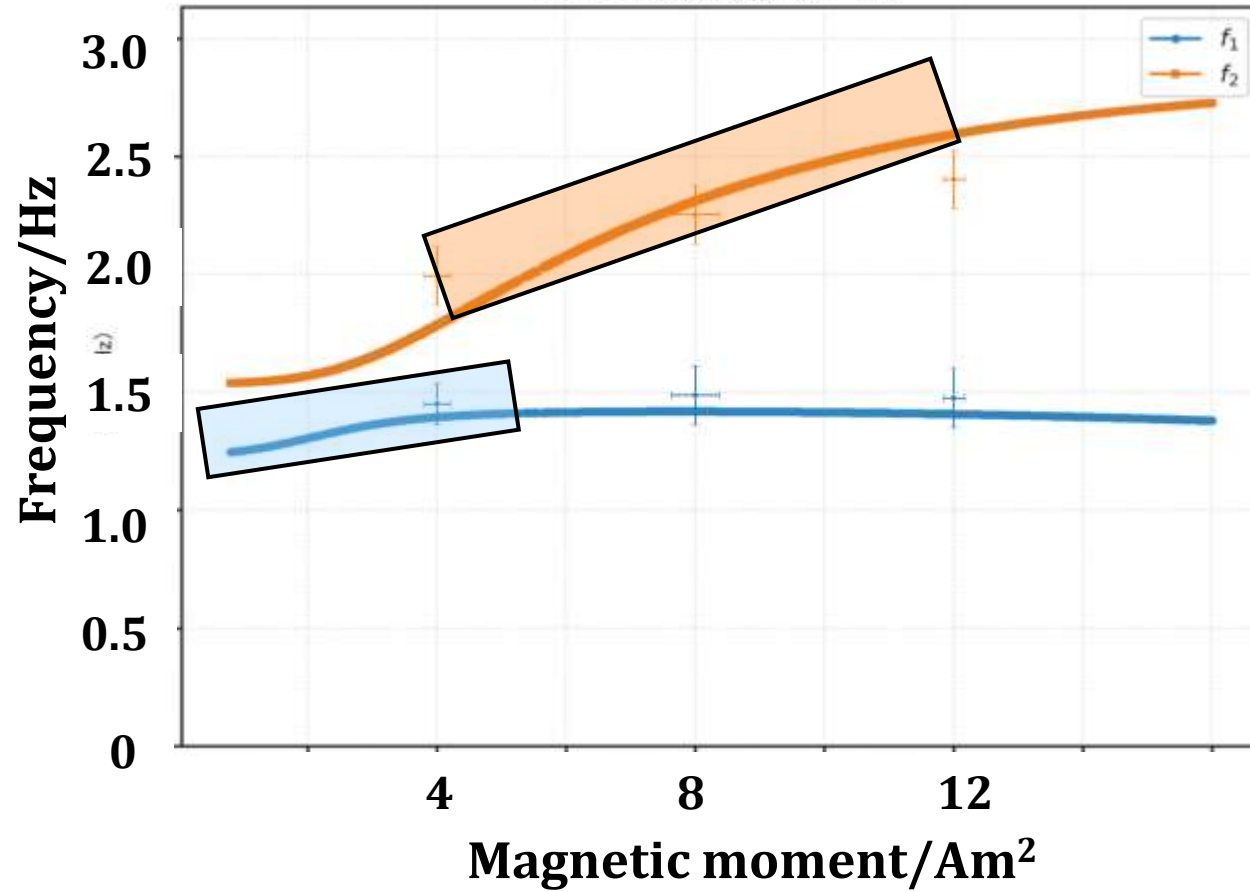
Magnetic coupling effect weakens

Suspension distance has small influence on gravitational potential energy

Parameter Analysis

Magnetic moment

Magnetic moment's influence on 1st 2nd frequencies



Magnetic coupling effect enhances

Gravitational-magnetic coupling

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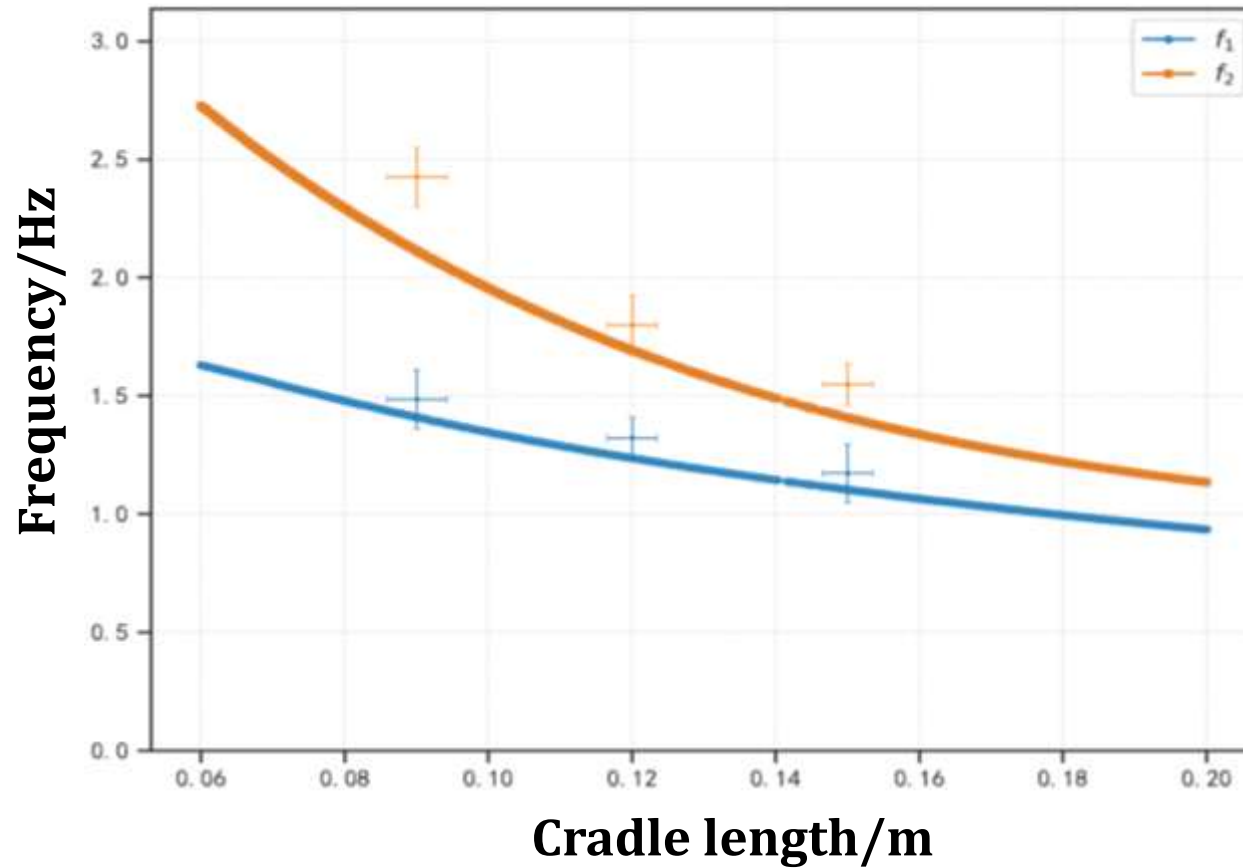
Theoretical Analysis

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Parameter Analysis

Cradle length



- Frequency decreases nonlinearly with increasing string length
- two frequency curves tend to converge in the long-string region

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Summary

Physical nature

dynamic coupling of **magnetic and gravitational forces**

Well-aligned

Theoretical analysis

Experiment

Kinematic analysis

Chaotic

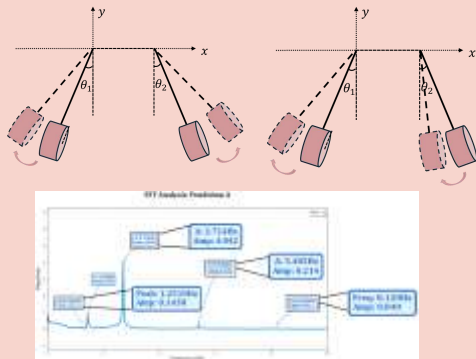
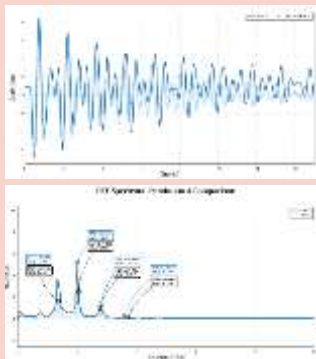
Multi-variable

Multi-periodic motion

Eigen-frequency

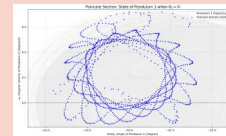
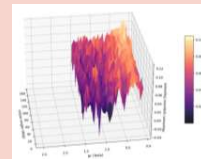
Non-linear behavior

Kinematic characteristics



MLE

Poincaré Section



Non/weak chaotic

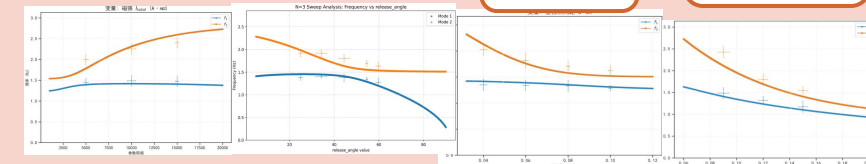
Number of cradle

Magnetic moment

Releasing angle

Suspension distance

Cradle length



1st dominated by gravitational, 2nd by magnetic potential energy

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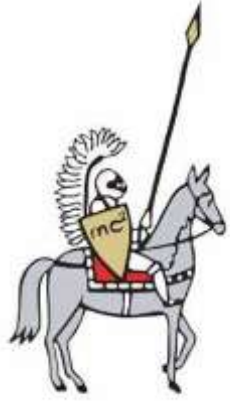
Conclusion

Error Analysis

- Parameter Measurement: Magnetization error 3.6%, damping coefficient error 2.8%, length uncertainty $\pm 0.0005\text{m}$, angular uncertainty $\pm 0.1\%$.
- Data Processing: Relative FFT resolution 1–3%.
- Modeling:
 1. Linear damping approximation: 0.1% – 0.5%
 2. Potential inconsistency in magnetic moments of the magnets
 3. Inability to ensure perfect coplanarity and absence of transverse perturbations: 0.1% – 1.2%

Overview

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15. Magnetic Newton's Cradle

Reporter Wanying Hu Beijing 101 High School



Thanks for Listening!



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Newly Added

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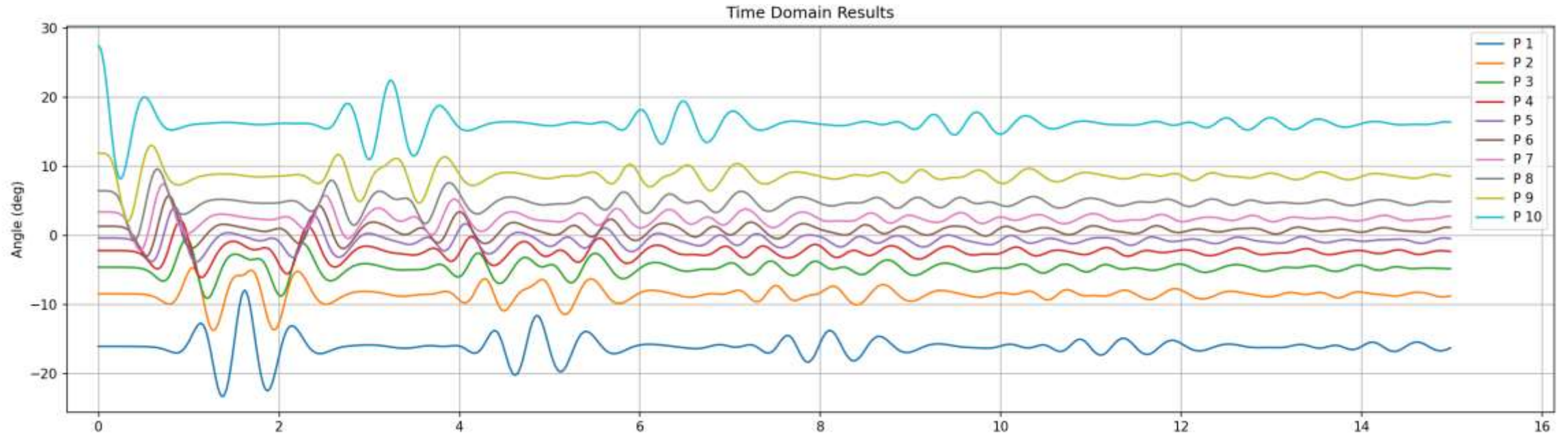
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“最大振幅”传递



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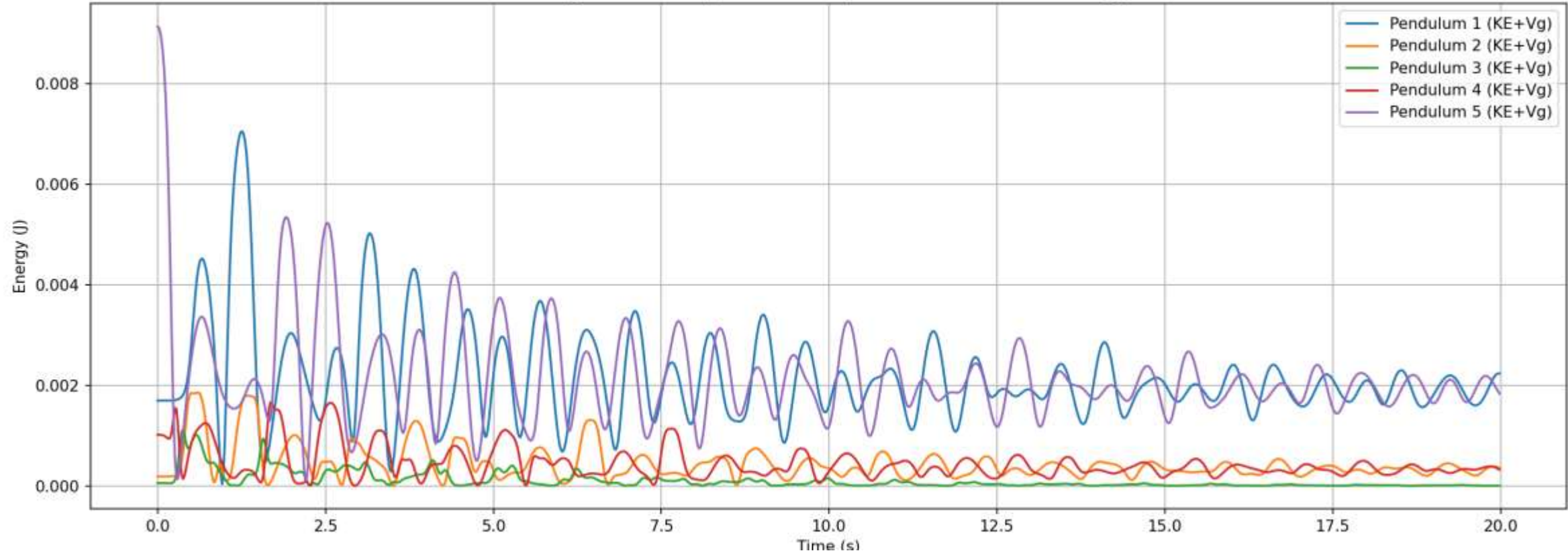
Theoretical Analysis

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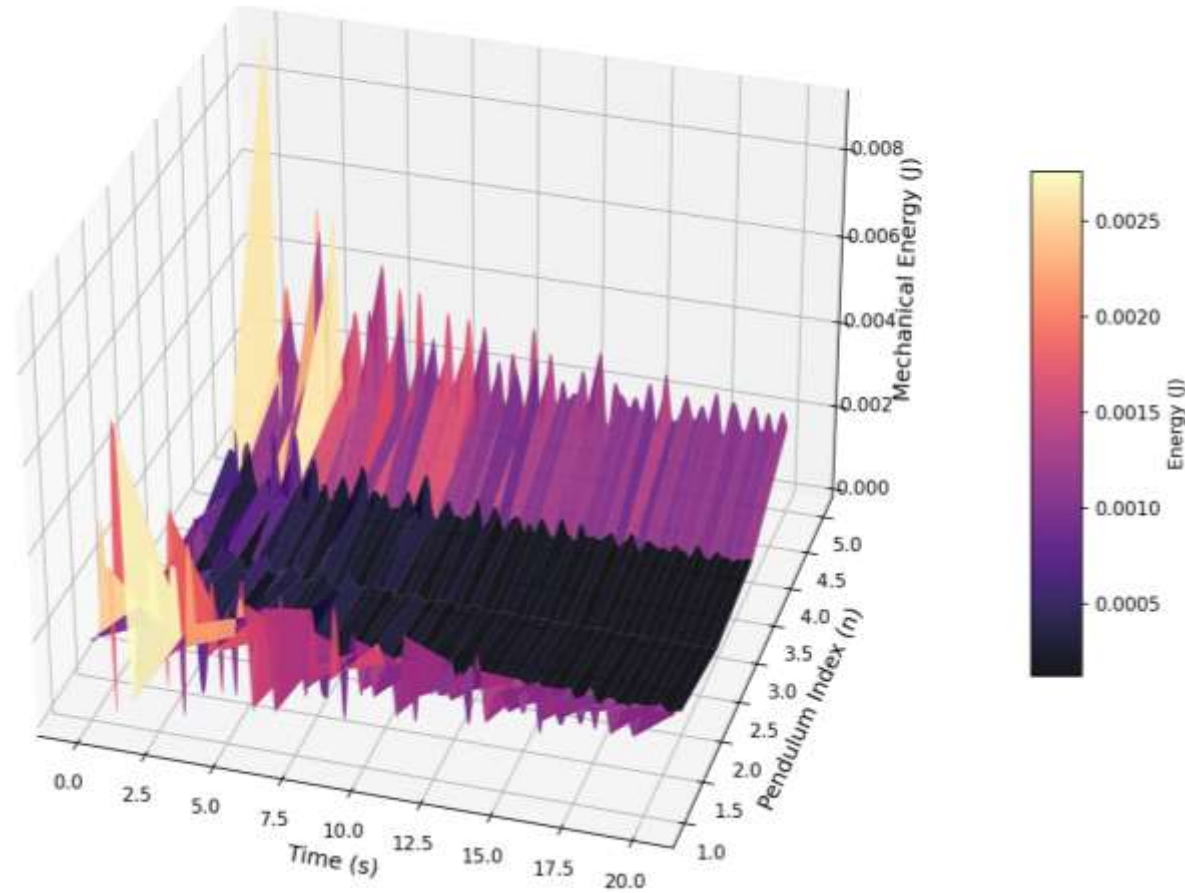
能量传递

Energy Distribution per Pendulum (Individual Mechanical Energy)



能量传递

Spatiotemporal Energy Distribution (Individual Mechanical Energy)



Introduction

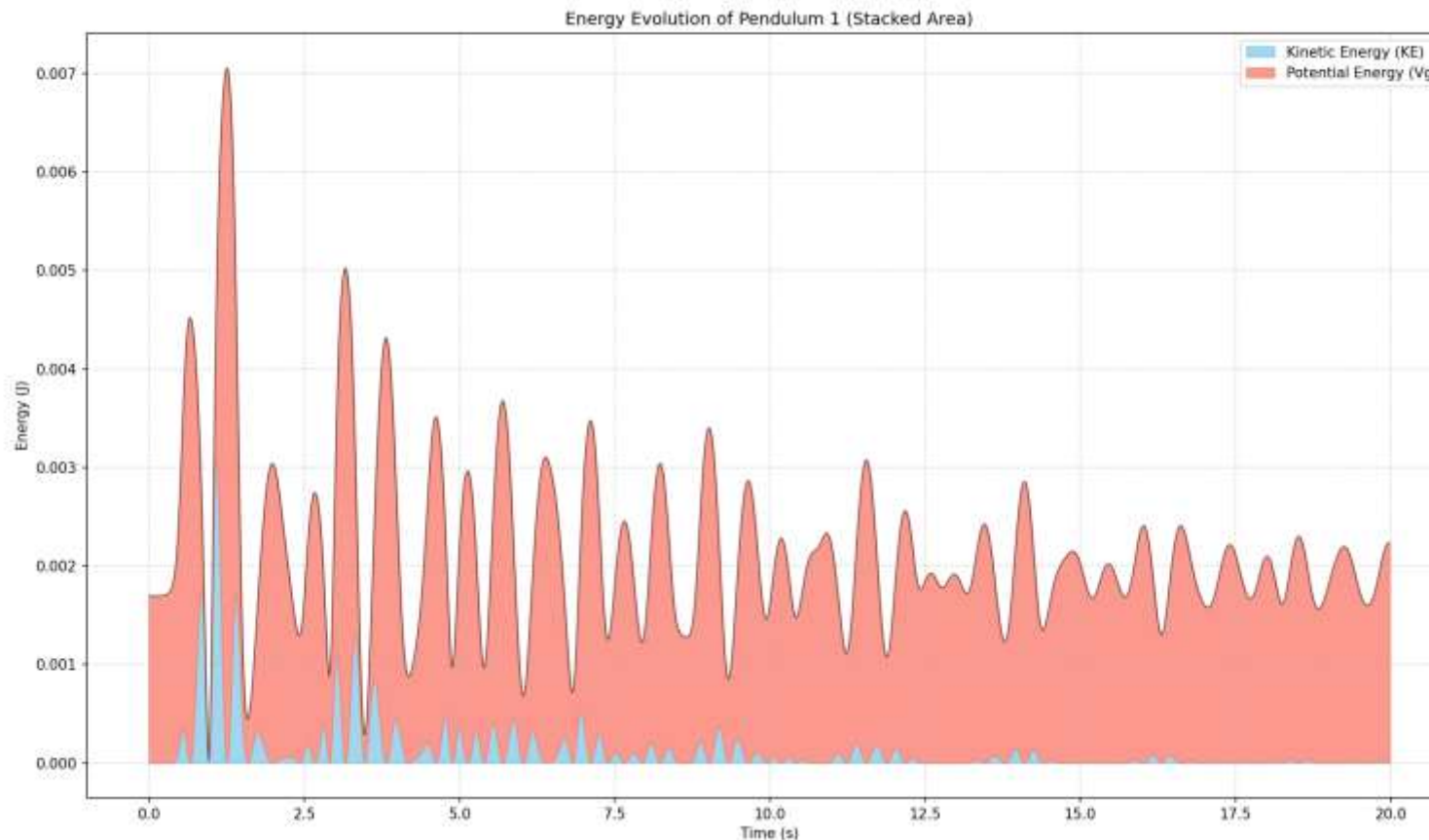
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动能重力势能对比—势能占主导



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色散关系

色散关系

1. 传播许可：通带与禁带 (Passbands & Bandgaps)

色散关系最直观的意义是划定了频率的“势力范围”。 ⌚+3

- 通带：色散曲线所在的频率区间，波动可以自由传播。 ⌚+1
- 禁带 (Bandgap)：曲线之外的频率区域。 ⌚+2
 - 高频禁带：由于晶格的离散性（原子之间有间距 a ），波长不能无限短。当 $K = \pi/a$ 时，波动会发生布拉格反射，形成驻波，能量无法向外传递。 ⌚
 - 低频禁带 (Gap)：在你的磁摆系统中，由于重力的存在，即使 $K \rightarrow 0$ ，频率 ω 也不为零。这意味着低于这个“门槛频率”的振动完全无法传播，只能在原地衰减。 ⌚+4

2. 能量流动的速度：群速度 (Group Velocity)

在色散曲线上，任意一点的斜率代表了能量传播的真实速度。 ⌚+2

- 群速度公式： $v_g = \frac{d\omega}{dK}$ 。 ⌚+3
- 物理意义：
 - 如果色散曲线是一条直线（如长波极限下的声波），所有频率跑得一样快，波形不会失真。 ⌚
 - 如果曲线是弯曲的，不同频率的成分会跑散。这就是为什么叫“色散”——原本聚在一起的波包会随着时间的流逝而“散架”（展宽）。 ⌚
 - 在布里渊区边界 ($K = \pi/a$)，斜率为零，意味着群速度为零，能量被锁死在原地。 ⌚+2

色散关系

Describing Function Method

1. 预测波的传播速度：群速度 (v_g)

在你的磁摆链里，一个脉冲能量跑得有多快？这不取决于 ω ，而取决于群速度。

- 计算方法：对色散公式求导 $v_g = \frac{d\omega}{dk}$ 。
- 分析点：
 - 长波极限 ($k \rightarrow 0$)：斜率最大，对应系统中波动的“声速”。
 - 布里渊区边界 ($k = \pi/a$)：斜率为 0。这意味着高频波动会被“困”在原地，变成驻波跑不出去。
 - 文献关联：文献中在测量孤子速度时，会先对比线性波的群速度。如果你的波跑得比最高群速度还快，那它极大概率是一个非线性的超声速孤子。

2. 解释“禁带”：为什么有些频率传不过去？

你在 FFT 频谱图中看到的那些“死区”，正是色散关系预言的。

- 低频禁带 (Lower Gap)：由 Ω_0 决定。如果你的驱动频率 $\omega < \Omega_0$ ，波动会以指数级速度衰减 (Evanescent wave)。
- 高频禁带 (Upper Gap)：由 ω_{max} 决定。
- 分析点：你可以改变磁铁间距 a ，观察禁带是如何上下移动的。这种通过调节物理结构来控制频率通过的能力，就是现在最前沿的声子晶体 (Phononic Crystals) 的研究核心。

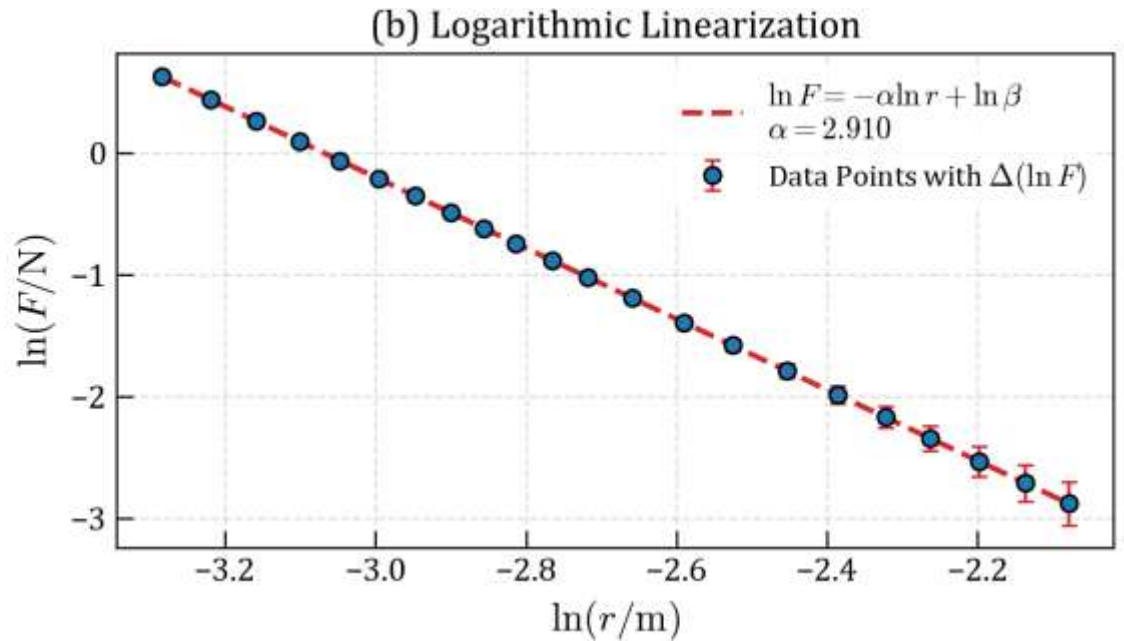
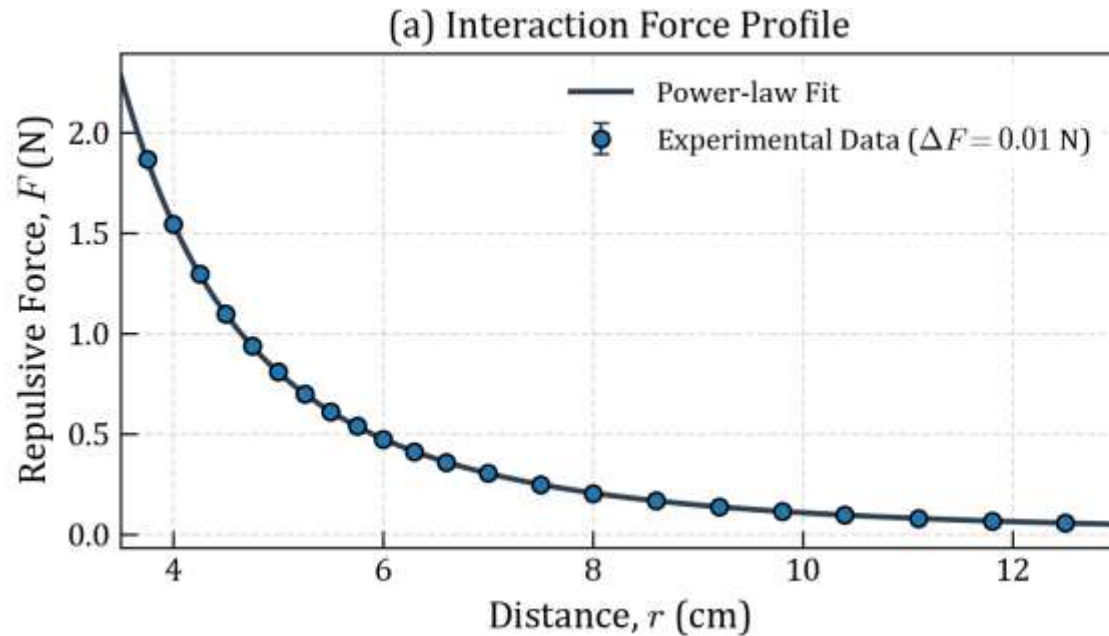
磁力幂律关系

磁场拟合结果: $B(z) = 1.6364e-05 * z^{(-1.910)}$

排斥力转换结果 $F(r) = \beta * r^{(-\alpha)}$:

α (衰减指数) = 2.910

β (幅值系数) = $1.3253e-04 \text{ N} \cdot \text{m}^\alpha$



色散与主频

1. 理论与现实的博弈（色散曲线的实部与虚部）

注意图中最上方的两对坐标轴，它们分别是 $Re(k)$ （实部）和 $Im(k)$ （虚部）。

- **实部 $Re(k)$ (左侧图)**: 这就是你刚理解的色散关系。它描述的是波的相位变化。当频率 f 在 Ω_0 和 ω_{max} 之间时，波可以传播。
- **虚部 $Im(k)$ (右侧图)**: 这是作者的高明之处。当频率掉进“禁带”（低于 Ω_0 或高于 ω_{max} ）时，波不能传播，而是会指数级衰减。这个虚部 $Im(k)$ 描述的就是衰减得有多快。
- **你的机会**: 你可以通过公式反解出 k 。如果频率在禁带里， k 会变成复数。画出这个虚部，就能解释为什么你的实验中高频振动传不到末端。

2. 非线性的“证据”：二倍频 (Second Harmonic)

图中特意用红色标出了 “First harmonic” 和 “Second harmonic”。

- 由于磁力和重力的非线性，当你以频率 ω 驱动系统时，系统内部会自动产生 2ω 甚至 3ω 的成分。
- **作者想说明**: 基频 (f) 可能在通带里传得很欢，但产生的二倍频 ($2f$) 可能已经超过了截止频率 ω_{max} ，掉进了禁带。所以你在 3D 频谱图中可以看到，基频能传很远，而高频成分在起始几个摆就消失了。

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释放驱动

2. 如果你是“释放驱动” (Pulse Excitation —— 你的实验情况)

你把磁摆拉到 50° 然后松手。这里没有持续的摇晃，那“速度”在哪？

答案隐藏在“松手”这个动作的时间尺度里：

- 瞬时释放 = 无限种频率：

在数学（傅里叶变换）中，一个突然的动作（位移从 50° 瞬间变为自由状态）等同于你在那一瞬间，以所有可能的频率同时摇晃了系统一遍。

- 能量的分配：

虽然你注入了所有频率，但能量并不是平分的。

- 你的释放动作越“干脆”，注入的高频成分就越多。
- 你的释放角度 θ_0 越大，注入的总能量就越高。

释放驱动激发的驻波——本质上是简正模式

数学边界条件

假设你的链条有 N 个磁摆，编号为 $n = 1, 2, \dots, N$ 。

在“自由-自由”边界下，第 1 个摆左边没有邻居，第 N 个摆右边没有邻居。

为了让系统的通用方程在边缘也能成立，物理学家使用了一个数学技巧——引入**虚拟磁摆** θ_0 和 θ_{N+1} ：

- 第 1 个摆不受左边受力，等价于令： $\theta_0 = \theta_1$
- 第 N 个摆不受右边受力，等价于令： $\theta_{N+1} = \theta_N$

$$\theta_n = A \cos \left[k \left(n - \frac{1}{2} \right) a \right]$$

释放驱动激发的驻波——边界条件

左边界

$$\theta_0 = A \cos \left[k \left(0 - \frac{1}{2} \right) a \right] = A \cos \left(-\frac{ka}{2} \right) = A \cos \left(\frac{ka}{2} \right)$$

$$\theta_1 = A \cos \left[k \left(1 - \frac{1}{2} \right) a \right] = A \cos \left(\frac{ka}{2} \right)$$

右边界

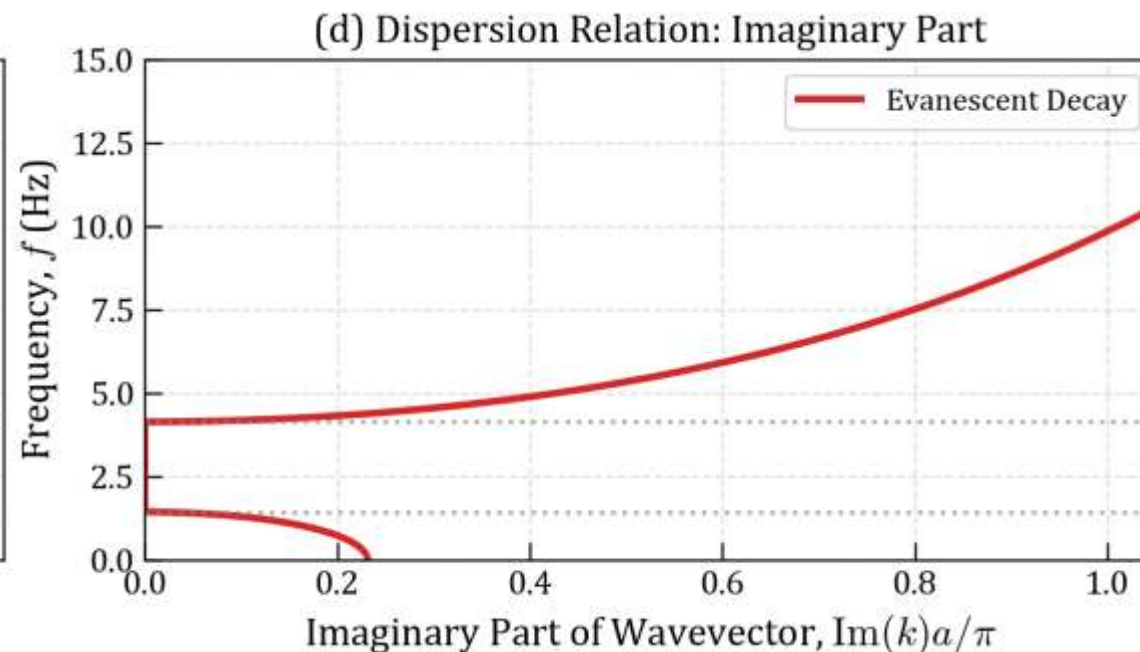
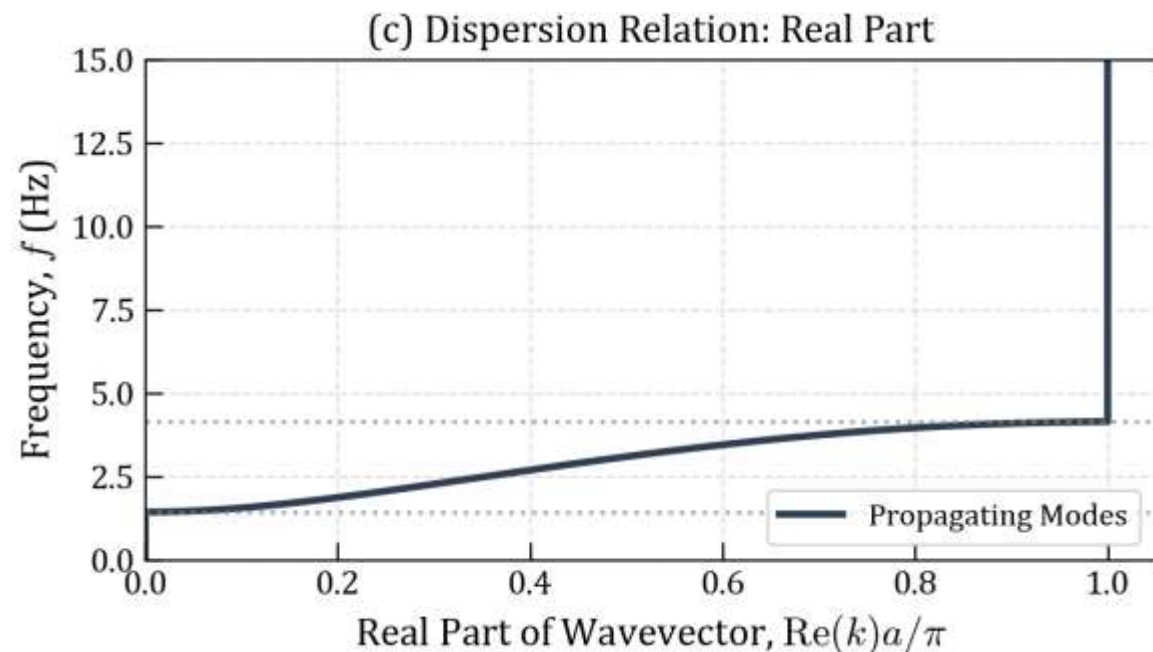
$$\cos \left[k \left(N + 1 - \frac{1}{2} \right) a \right] = \cos \left[k \left(N - \frac{1}{2} \right) a \right] \quad \sin \left(\frac{ka}{2} \right) = 0 \rightarrow k = 0 \text{ (需要保证在第一布里渊区中)}$$

$$-2 \sin(kNa) \sin \left(\frac{ka}{2} \right) = 0$$

$$\sin(kNa) = 0 \rightarrow kNa = m\pi \quad (m = 1, 2, 3, \dots)$$

$$k_m = \frac{m\pi}{Na} \quad m = 0, 1, 2, \dots, N-1$$

色散关系求解

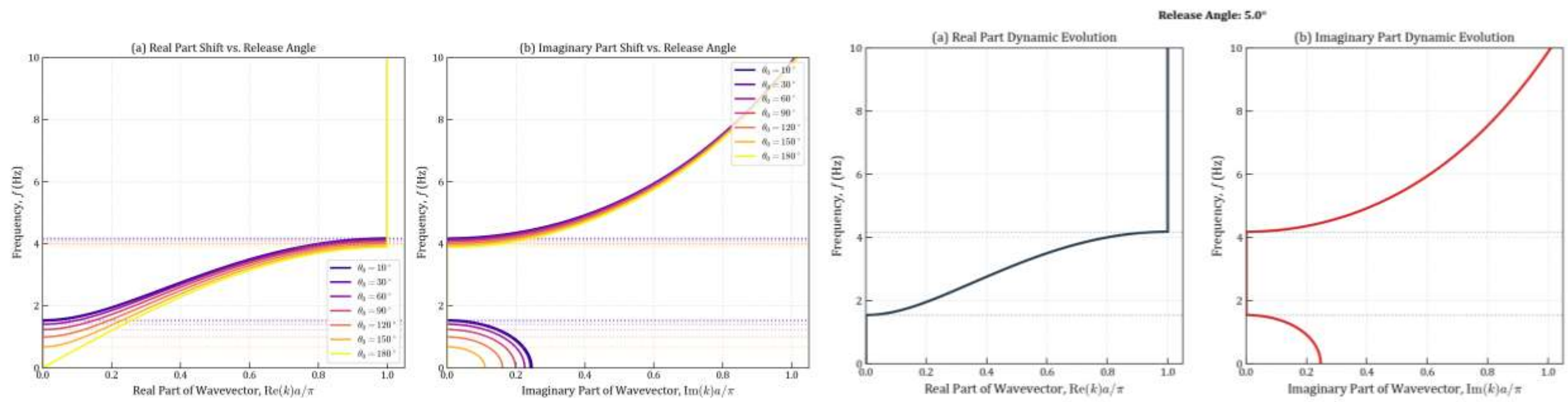


===== 色散关系物理特征量 =====

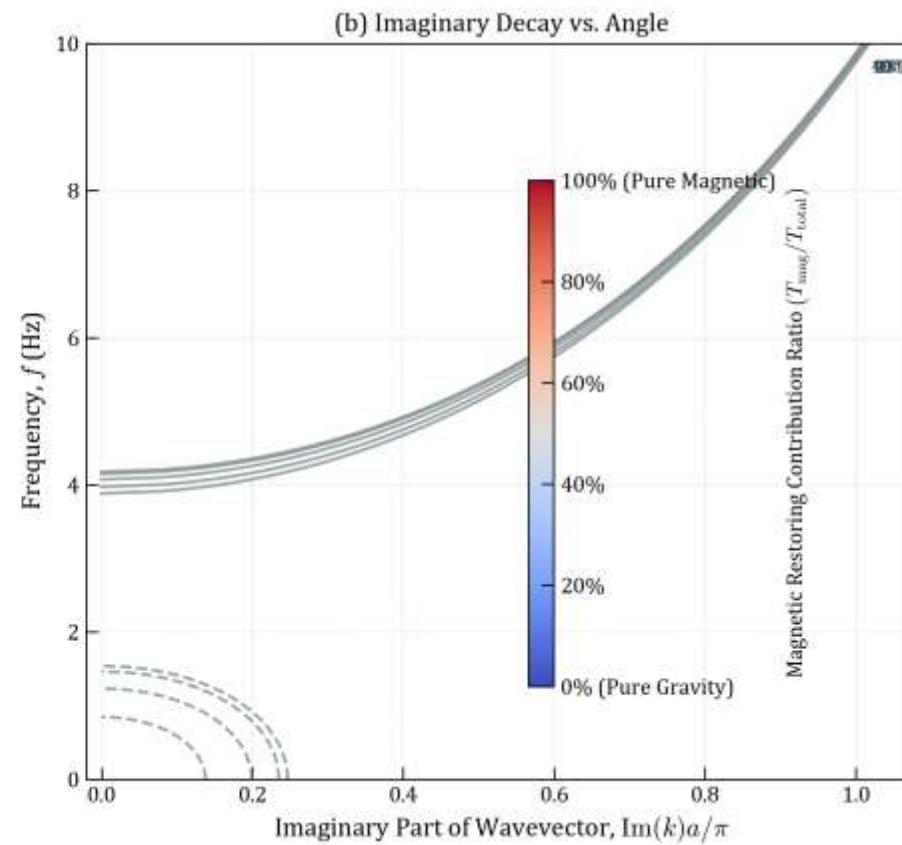
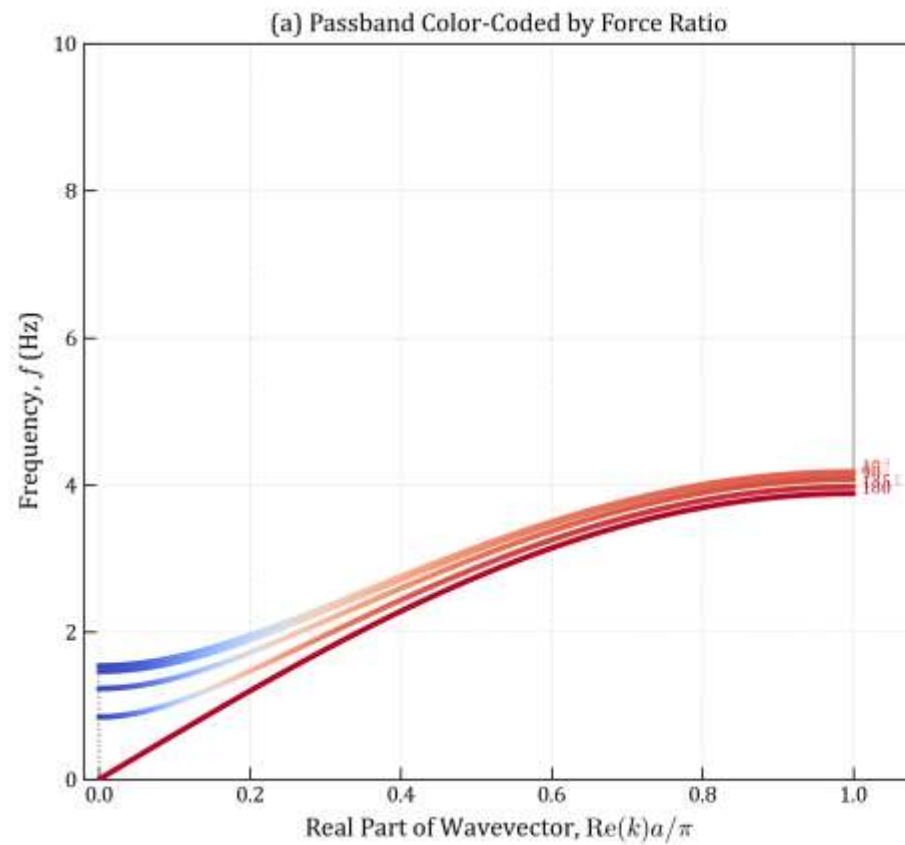
重力下载截止频率 $f_0 = 1.444$ Hz

最高截止频率 $f_{\text{max}} = 4.144$ Hz

色散关系-释放角度



色散关系-磁和重力



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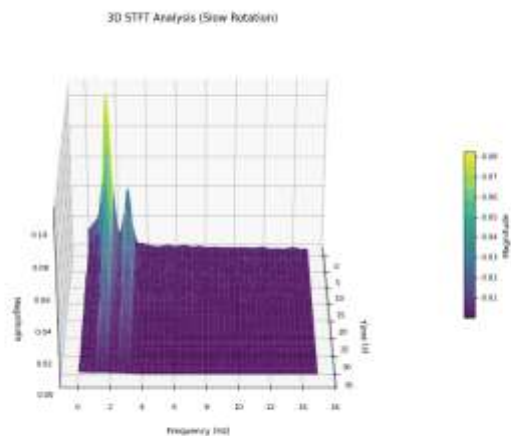
Experiment

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色散关系

1. 看一下其他变量对色散关系的影响
2. 根据几何设置 把系统具有的频率标注在图中，并做实验验证

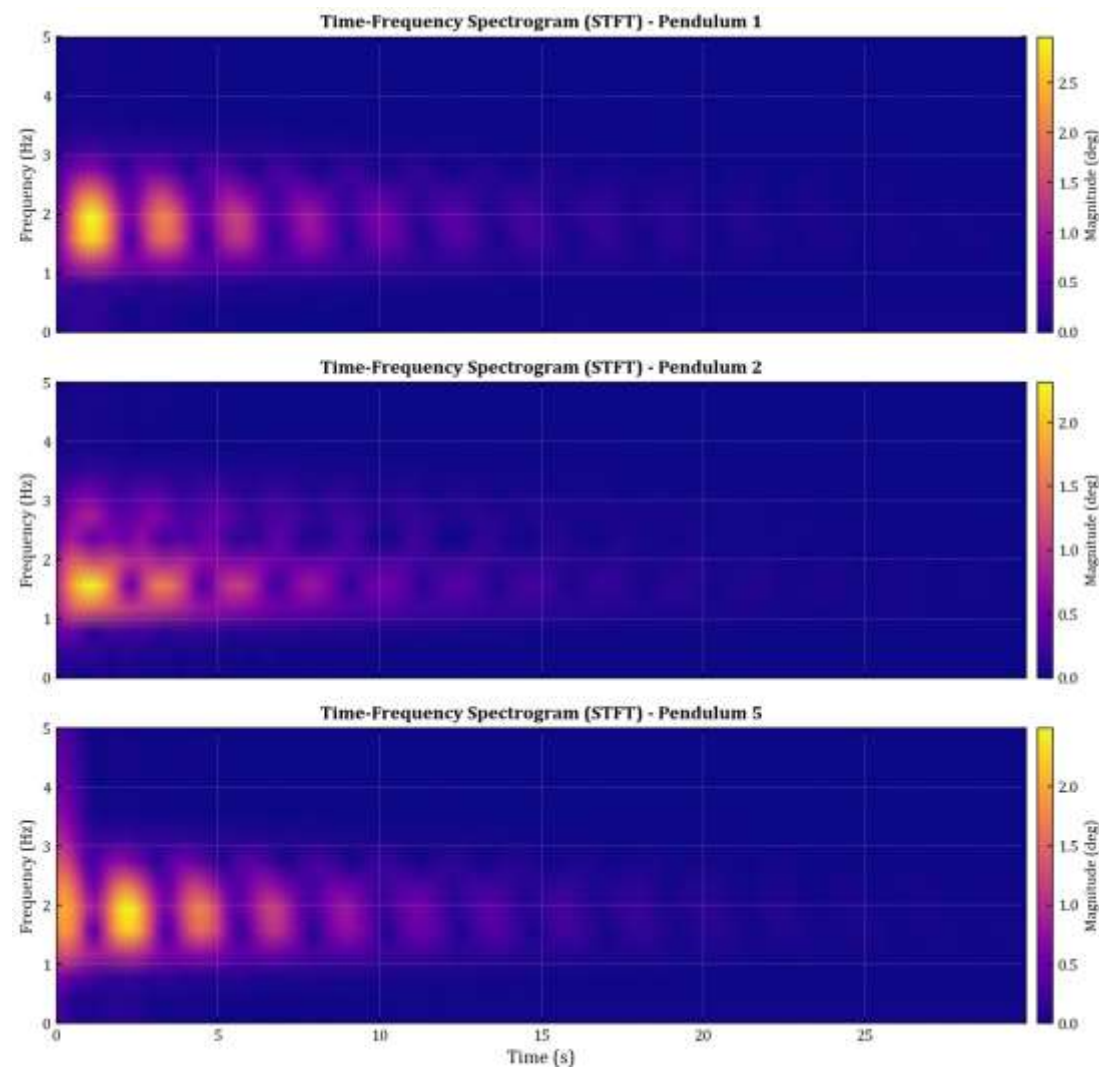
不同主导频率随时间的变化



多给实验数据做几组stft，研究频率随着时间的变化

文献中研究的是随着位置的变化，那这里可以研究随着时间的变化

不同主导频率随时间的变化



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能量角度: *kink and phonon*

能量和频率的关系?

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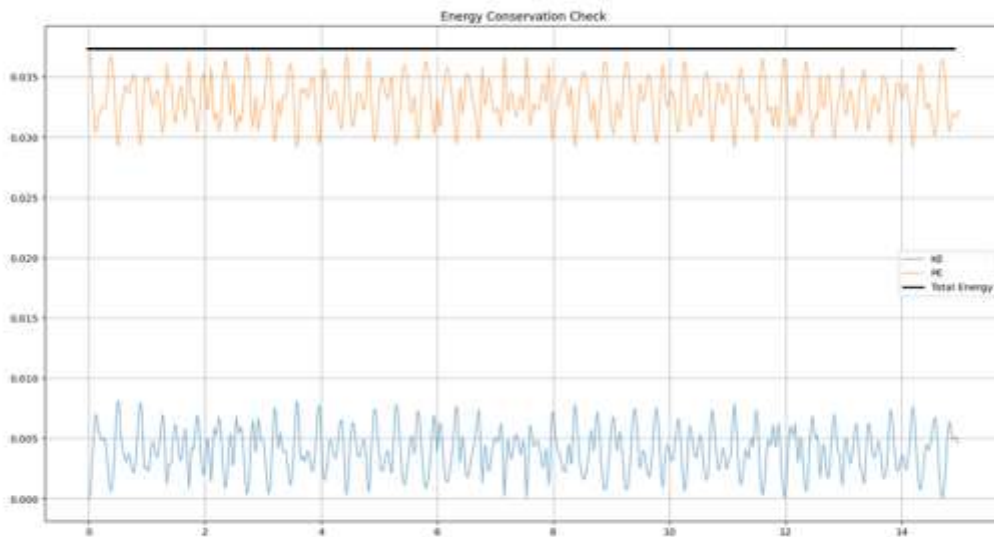
Theoretical Analysis

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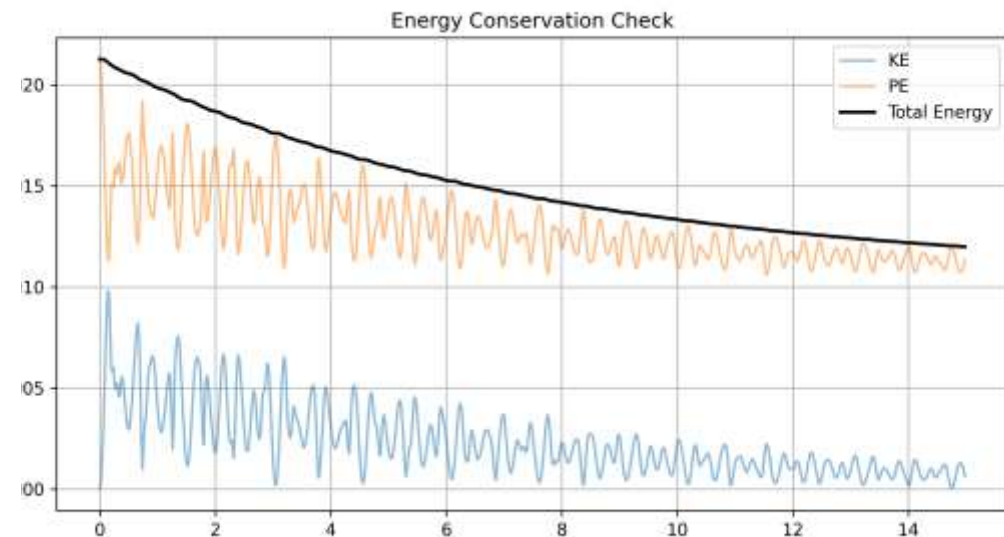
Conclusion

Conservation of Energy

When the damping coefficient is zero, the simulation results show that the total energy remains constant, which is consistent with the law of **conservation of energy**

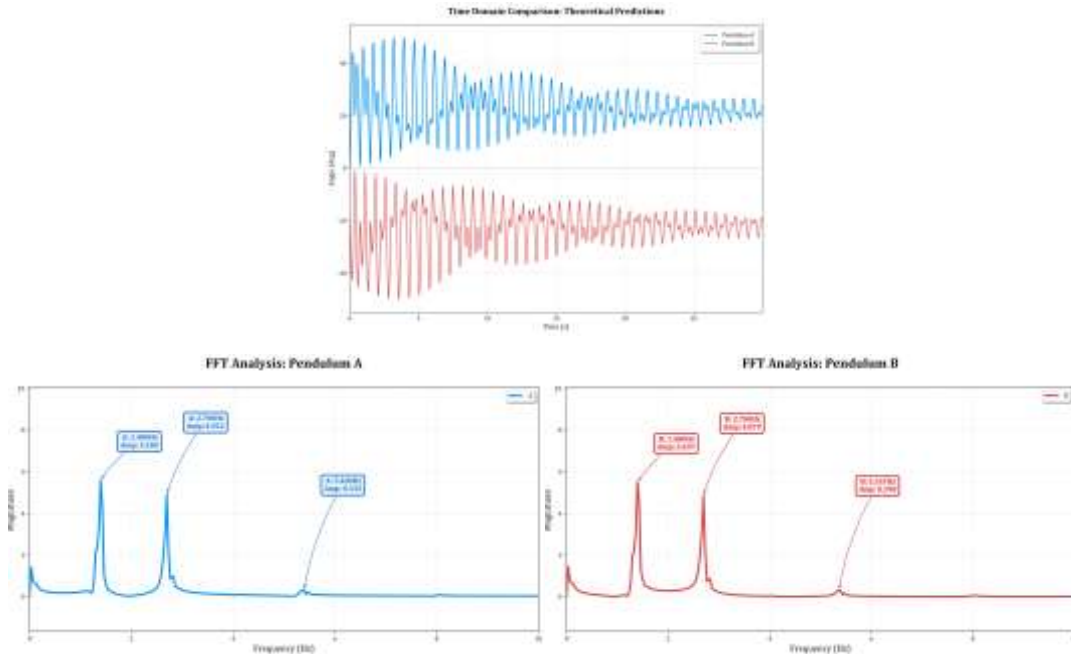


When the damping coefficient is non-zero, the total energy **decays exponentially** and eventually levels off at a **non-zero constant**.



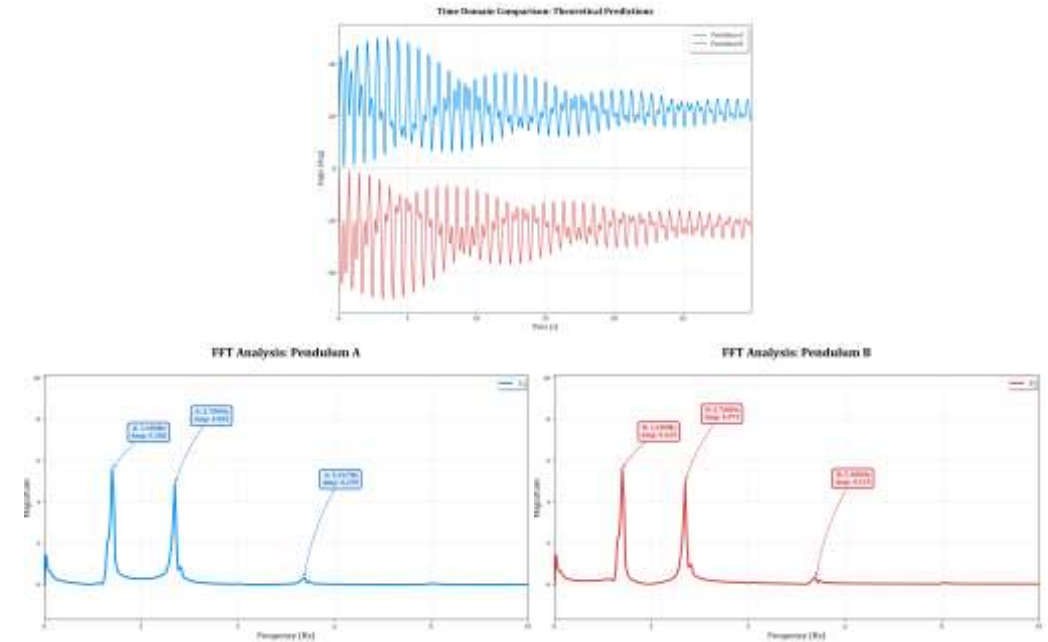
Symmetry Verification

Releasing angle for cradle 1: 20° , cradle 2: 0°



dominant frequency is 1.400 (Amp=5.497) ,
2.700Hz (Amp=4.879)

Releasing angle of cradle 1: 0° , cradle 2: 20°



dominant frequency is 1.400 and 2.700Hz

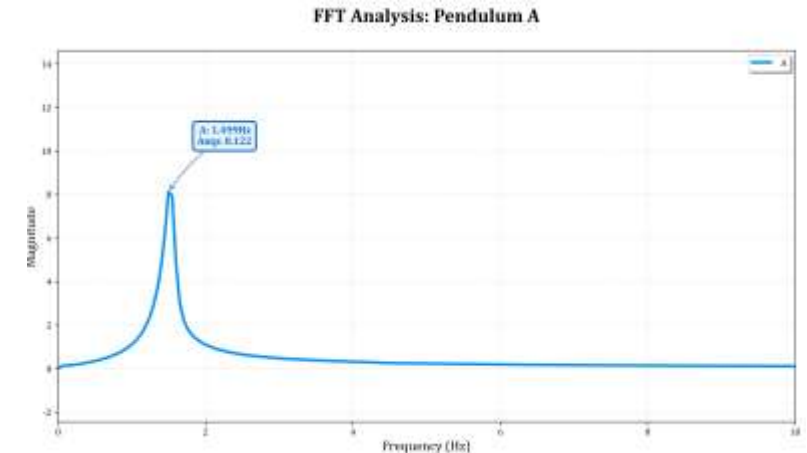
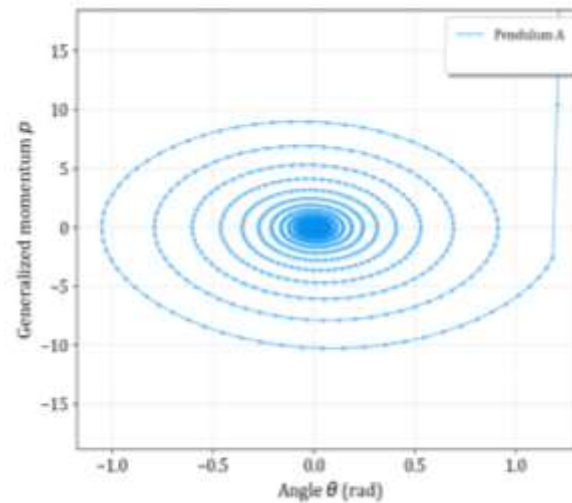
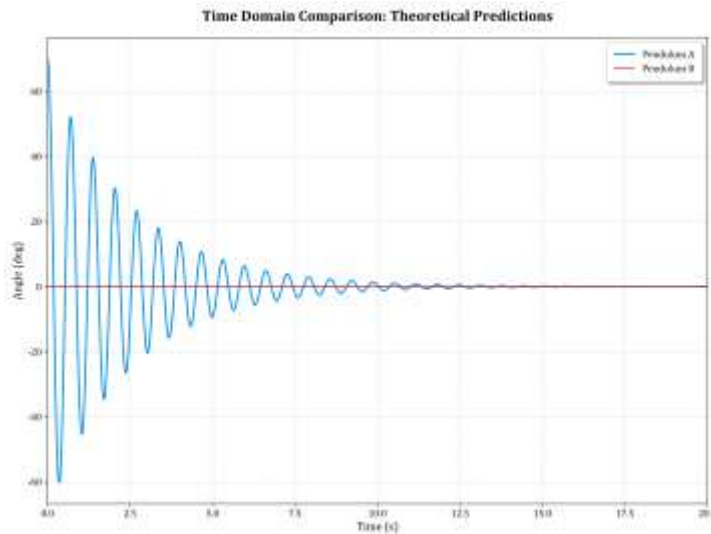
Limit Test

$$s \rightarrow \infty$$

$$s \rightarrow \infty \quad r_{ij} \rightarrow \infty \quad \vec{B}(\vec{r}) \rightarrow 0 \quad V_{mag} \rightarrow 0 \quad L = T - V = \frac{1}{2} I_{eq} \sum_{i=1}^n \dot{\theta}_i^2 - \sum_{i=1}^n M g (l + R) (1 - \cos \theta_i)$$

$$I_{eq} \ddot{\theta}_k + \frac{\partial V(\theta)}{\partial \theta_k} + \gamma \dot{\theta}_k = 0 \rightarrow I_{eq} \ddot{\theta}_k + \gamma \dot{\theta}_k + M g (l + R) \sin \theta_k = 0$$

Magnetic interaction vanishes, the system degenerates into multiple independent compound pendulums



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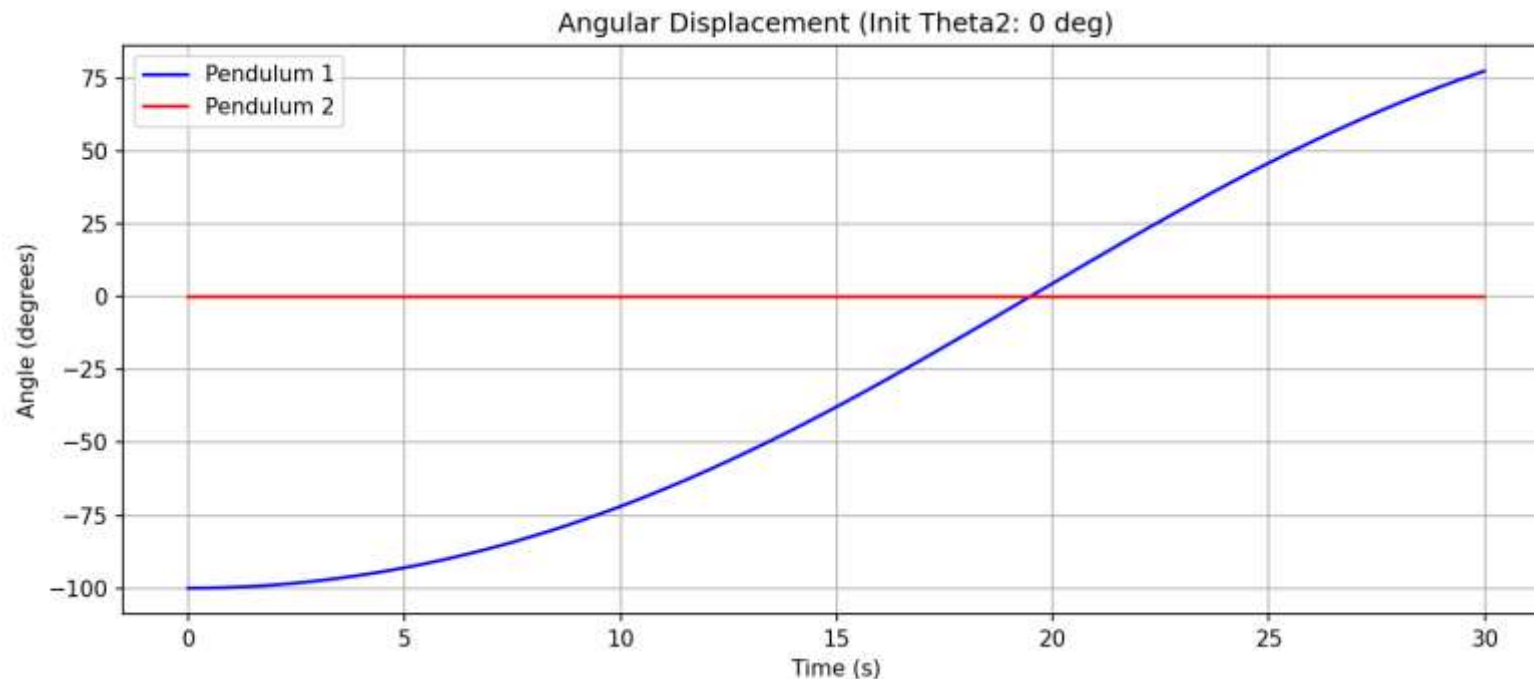
Conclusion

Limit Test

$$l \rightarrow \infty$$

$$l \rightarrow \infty \quad \theta_i = \arcsin\left(\frac{x_i}{l+R}\right) \rightarrow 0 \quad \sin\theta_i \approx \theta_i \quad \cos\theta_i \approx 1 - \frac{\theta_i^2}{2} \quad V_{grav} = Mg(l+R)(1 - \cos\theta_i) \rightarrow 0$$

Gravity no longer provide an effective restoring force, effectively placing the system in a 'zero-gravity' environment



Normal Modes

Solve for the eigenvalues of this matrix:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

$$\det(\lambda^2 \mathbf{M} + \lambda \mathbf{D} + \mathbf{K}) = 0$$

$$\mathbf{A} = \begin{bmatrix} 0 & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{D} \end{bmatrix}$$

For a system consisting of N magnetic pendulums, $2N$ complex eigenvalues are obtained.

$$\lambda = \sigma \pm i\omega_d$$

Decay rate: contributed by damping

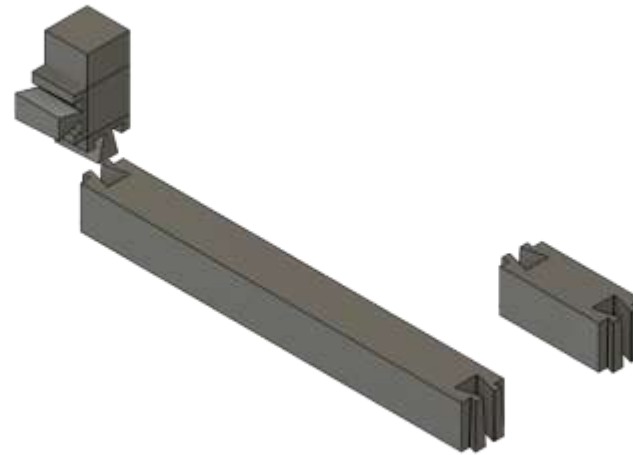
Oscillation frequency

Experimental Setup

3D-printed PETG bracket



Bracket top: Suspends magnetic pendulums and controls intervals.



Bracket connection: Fixes bracket height; non-ferromagnetic.

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cradle



Bearings: Free rotation and friction reduction

Magnetic pendulum arm:
Controls pendulum length

Magnetic pendulum basket: Houses the magnetic pendulum and prevents flipping

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protractor



Protractor: Controls release angle

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Matrix Representation

$$I_{eq}\ddot{\theta}_k + \gamma\dot{\theta}_k + \frac{\partial V(\theta)}{\partial \theta_k} = 0$$

The n coupled equations for all magnetic pendulums in the system can be formulated in matrix form as

$$\mathbf{M}\ddot{\boldsymbol{\theta}} + \mathbf{D}\dot{\boldsymbol{\theta}} + \mathbf{K}\boldsymbol{\theta} = 0$$

Where M is the mass matrix: $\mathbf{M} = \begin{bmatrix} I_{eq} & 0 & \dots \\ 0 & I_{eq} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} = I_{eq} \cdot \mathbf{I}_{n \times n}$

inertial resistance to changes in the system's state of motion

Dis the damping matrix: $\mathbf{D} = \begin{bmatrix} \gamma & 0 & \dots \\ 0 & \gamma & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} = \gamma \cdot \mathbf{I}_{n \times n}$

influence of external damping

K is the stiffness matrix: $\mathbf{K}: K_{ij} = \frac{\partial^2 V}{\partial \theta_i \partial \theta_j}$

restoring torque acting on the magnetic pendulums when deviating from equilibrium

In a damped system, energy dissipates, and the system exhibits complex modes.

To theoretically solve for the damped normal modes, the equations of motion must be transformed into state-space form.

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State-Space

Fundamentally, this involves transforming a system of second-order differential equations into a system of first-order equations. First, a state vector X is constructed, containing all the information required to describe the system at any given instant.

For 3 cradles, position $\Theta = [\theta_1, \theta_2, \theta_3]^T$, angular velocity $\dot{\Theta} = [\omega_1, \omega_2, \omega_3]^T$.

$$X = \begin{bmatrix} \Theta \\ \dot{\Theta} \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

The original second-order system is then reformulated as a first-order equation with respect to X

$$\dot{X} = AX$$

State-Space

$$\dot{X} = AX$$

Construct the state matrix A

First, take the time derivative of X

$$\dot{X} = \begin{bmatrix} \dot{\Theta} \\ \ddot{\Theta} \end{bmatrix}$$

Next, solve for $\ddot{\Theta}$ from the original equation $M\ddot{\Theta} + D\dot{\Theta} + K\Theta$

$$\ddot{\Theta} = -M^{-1}K\Theta - M^{-1}D\dot{\Theta}$$

$$\begin{bmatrix} 0 & I \\ -M^{-1}K & -M^{-1}D \end{bmatrix}$$

Combine the velocity and acceleration components into a $2N \times 2N$ matrix

$$\underbrace{\begin{bmatrix} \dot{\Theta} \\ \ddot{\Theta} \end{bmatrix}}_{\dot{X}} = \underbrace{\begin{bmatrix} 0 & I \\ -M^{-1}K & -M^{-1}D \end{bmatrix}}_A \underbrace{\begin{bmatrix} \Theta \\ \dot{\Theta} \end{bmatrix}}_X$$

Normal Modes

The eigenvalues of the state matrix A represent the system's eigenfrequencies, and the corresponding eigenvectors reflect the amplitude ratios and phase relationships of the pendulums.

$$\det(A - \lambda I) = 0$$

$$\det(\lambda^2 \mathbf{M} + \lambda \mathbf{D} + \mathbf{K}) = 0$$

Expanding this determinant yields a polynomial of degree $2N$

Solving it results in $2N$ complex eigenvalues (which are essentially N pairs of conjugate eigenvalues)

$$\lambda = \sigma \pm i\omega_d$$

Normal Modes

$$\lambda = \sigma \pm i\omega_d$$

$$\theta(t) = C \cdot e^{\lambda t}$$

$$\theta(t) = C \cdot e^{(\sigma + i\omega)t} = C \cdot e^{\sigma t} \cdot e^{i\omega t}$$

using $e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$

$$\theta(t) = C \cdot e^{\sigma t} (\cos(\omega t) + i\sin(\omega t))$$

Decay rate: contributed by damping

$$\lambda = \sigma \pm i\omega_d$$

Oscillation frequency

Exponential function → envelope

$\sigma < 0$, system is stable

Periodic
function → oscillation
frequency

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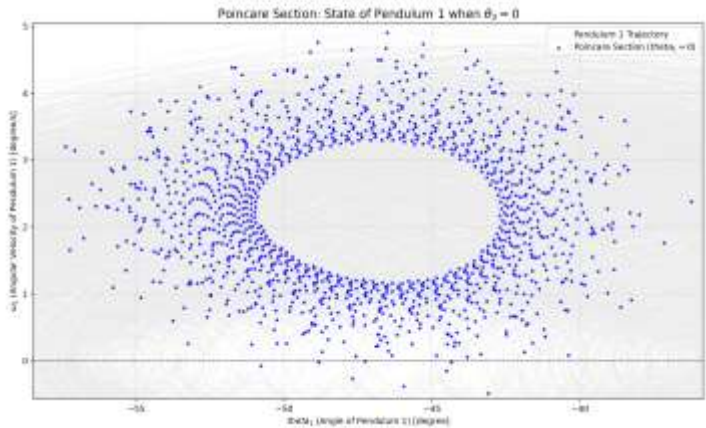
Conclusion

Poincaré Section

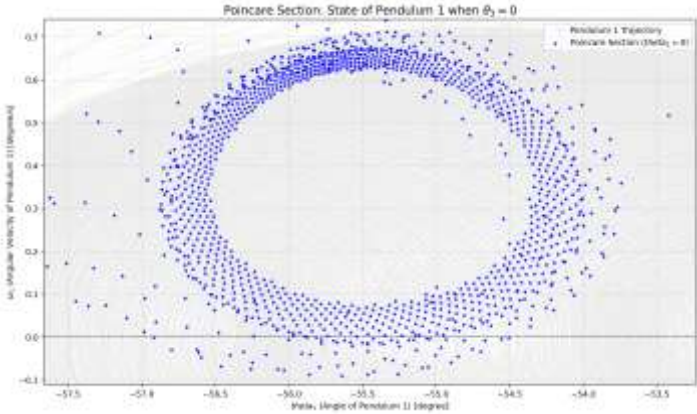
B_r

Poincaré section of cradle 1 at $\theta_3 = 0$ (5-cradle system)

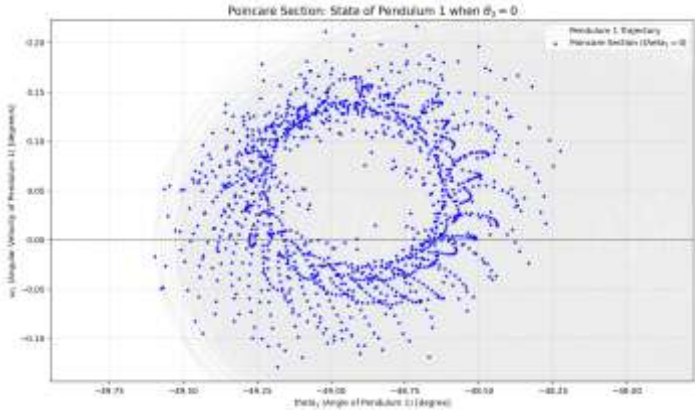
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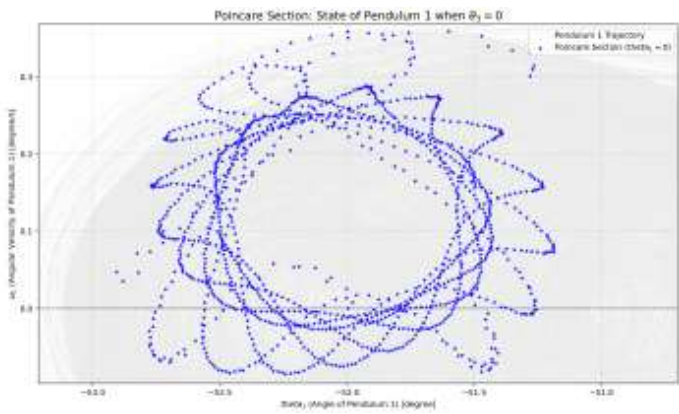
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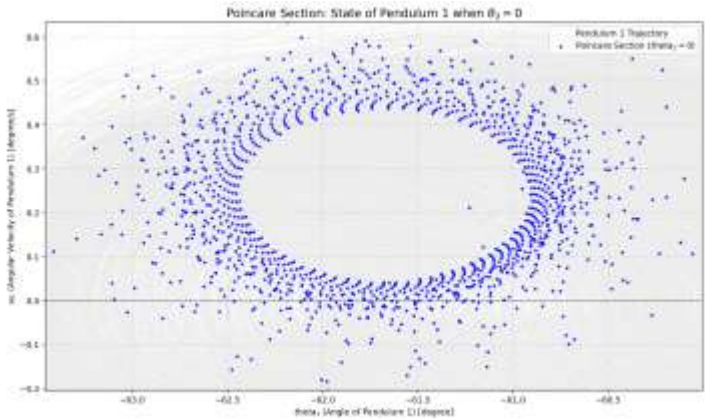
$I=8000$



$I=8700$



$I=11000$



Maximum Lyapunov Exponent

Benettin algorithm

$$\mathbf{y}_{shadow}(0) = \mathbf{y}_{ref}(0) + \mathbf{d}_0$$

Calculate the new distance between the two trajectories after one time step.

$$d_1 = \left| \mathbf{y}_{shadow}(\tau) - \mathbf{y}_{ref}(\tau) \right|$$

$$\mathbf{y}_{shadow}^{new} = \mathbf{y}_{ref} + d_0 \cdot \frac{(\mathbf{y}_{shadow} - \mathbf{y}_{ref})}{d_1}$$

After k renormalizations, the discrete estimation formula for the maximum Lyapunov exponent is:

$$\lambda \approx \frac{1}{k \cdot \tau} \sum_{i=1}^k \ln \left(\frac{d_i}{d_0} \right)$$

Error Analysis

$$\omega_d = \sqrt{\omega_0^2 - \left(\frac{\gamma}{2I_{\text{eq}}}\right)^2} = \sqrt{\omega_0^2 - \lambda^2}$$

$$\omega_d = \omega_0 \sqrt{1 - \frac{\lambda^2}{\omega_0^2}} \approx \omega_0 \left(1 - \frac{1}{2} \frac{\lambda^2}{\omega_0^2}\right)$$

$$\frac{\Delta\omega}{\omega_0} \approx \frac{1}{2} \frac{\lambda^2}{\omega_0^2} = \frac{1}{2} \left(\frac{\lambda}{2\pi f_0}\right)^2$$

Renolds Number

$$\rho: 1.2 \text{ kg/m}^3$$

$$\mu: 1.8 \times 10^{-5} \text{ Pa} \cdot \text{s}$$

$$D: 0.03 \text{ m}$$

$$l: 0.0892 \text{ m}$$

$$f: 1.454 \text{ Hz}$$

$$v = \omega \cdot l \cdot \sin(\theta)$$

$$v = (2 \times \pi \times 1.454) \times 0.0892 \times 0.342 \approx \mathbf{0.278 \text{ m/s}}$$

$$Re = \frac{\rho \cdot v \cdot D}{\mu}$$

$$Re = \frac{1.2 \times 0.278 \times 0.03}{1.8 \times 10^{-5}} = \frac{0.010008}{0.000018} \approx \mathbf{556}$$